

REPORT ON PROCEEDINGS BEFORE

PUBLIC ACCOUNTABILITY AND WORKS COMMITTEE

INQUIRY INTO DATA CENTRES

CORRECTED

At Macquarie Room, Parliament House, Sydney on Friday 29 May 2026

The Committee met at 09:00.

PRESENT

Ms Abigail Boyd (Chair)
The Hon. Mark Buttigieg
The Hon. Scott Farlow (Deputy Chair)
The Hon. Jacqui Munro
The Hon. Emily Suvaal

PRESENT VIA VIDEOCONFERENCE

The Hon. Taylor Martin

The CHAIR: Welcome to the fourth hearing of the Committee's inquiry into data centres. I acknowledge the Gadigal people of the Eora nation, the traditional custodians of the lands on which we are meeting today. I pay my respects to Elders past and present, and celebrate the diversity of Aboriginal peoples and their ongoing cultures and connections to the lands and waters of New South Wales. I also acknowledge and pay my respects to any Aboriginal and Torres Strait Islander people joining us today or watching us online. My name is Abigail Boyd. I am the Chair of the Committee.

I ask everyone in the room to please turn their mobile phones to silent. Parliamentary privilege applies to witnesses in relation to the evidence they give today. However, it does not apply to what witnesses say outside of the hearing. I urge witnesses to be careful about making comments to the media or to others after completing their evidence. In addition, the Legislative Council has adopted rules to provide procedural fairness for inquiry participants. I encourage Committee members and witnesses to be mindful of these procedures.

Mr COLIN CRISAFULLI, General Manager, Future Grid and Asset Management, Endeavour Energy, sworn and examined

Ms FATIMA BAZZI, Group Executive, Customer, Ausgrid, sworn and examined

Ms ANNIE PEARSON, Chief Customer and Corporate Affairs Officer, Essential Energy, sworn and examined

The CHAIR: I welcome our first panel of witnesses. Thank you very much for making the time to give evidence. Would you like to commence with a short opening statement?

FATIMA BAZZI: Chair, members of the Committee, thank you for the opportunity to appear today on behalf of Ausgrid. My name is Fatima Bazzi and I am the Group Executive, Customer. I'm pleased to be here today with my fellow New South Wales distribution networks, Endeavour Energy and Essential Energy. Together, we own and operate the electricity distribution networks for New South Wales, powering the lives of around seven million people. Ausgrid serves the Sydney CBD, the Central Coast and the Hunter regions, keeping the lights on for over four million people across more than 22,000 square kilometres. We address the Committee together today with a shared interest in ensuring data centre customer growth is planned for, managed and located to maximise the benefits and mitigate risks to our networks and the customers they serve.

Distribution networks are at the heart of the energy transition, delivering infrastructure needed for a cleaner, more reliable and affordable energy system. The grid is a shared asset and the costs of the transition are shared. We are delivering initiatives such as community batteries, kerbside EV charging and renewable energy zones to ensure everyone, regardless of their circumstances, shares the benefits of the transition. The rapid growth of data centre customers presents one of the most significant challenges and opportunities facing our network in a generation. Today they account for approximately 4 per cent of operational load in New South Wales. By 2040 we predict that to rise to between 17 and 30 per cent. That kind of growth in that time frame is unlike anything we have ever seen.

We welcome this inquiry's efforts to ensure that data centre customer growth does not come at the expense of affordable and reliable electricity or other priorities. We hear the real concerns about energy bills and power reliability, and we take that very seriously. But, done well, connecting more data centre customers can deliver very meaningful benefits. Increased demand improves network utilisation. This lowers the share each customer pays and reduces bills for everyone. This is a genuine opportunity to ease cost pressures on households and small businesses.

I draw the Committee's attention to three key themes from our joint submission, which shows how distribution networks are already working to realise these benefits and where more support is needed. First, in New South Wales, the customer who caused network costs pays for them. You will hear this referred to as "causer pays". We use a range of mechanisms to apply this principle. By design, existing customers are protected from cross-subsidisation. Ausgrid is proactively exploring how to leverage existing funding mechanisms to provide even stronger protections for our existing customer base.

Second, parts of the New South Wales distribution system already have capacity to connect new large loads with low lead time and risk. For example, the Hunter-Central Coast Renewable Energy Zone is underway. These regions are well placed to host data centre load close to new generation. We encourage the inquiry to consider a proactive strategy that identifies precincts where connections should be encouraged. Third, the rules we operate under were written for a different era. The five-year regulatory cycle was not designed for this pace of change. Networks must be able to respond with greater flexibility and make strategic, anticipatory investments to meet growth at lower costs. We urge the inquiry to consider reforms that support this. I'll now hand over to Colin from Endeavour Energy, who will talk through what this looks like on the ground in Western Sydney.

COLIN CRISAFULLI: Thank you, Chair and Committee, for the opportunity to speak today. Endeavour Energy services the Illawarra, Southern Highlands, Blue Mountains and Greater Western Sydney. Western Sydney is Australia's fastest growing region and home to Australia's first new sustainable city in 100 years, a new international airport and more than \$50 billion in committed infrastructure. We also acknowledge the traditional custodians of the land on which this growth is occurring, including the Dharug, Dharawal and Gandangara peoples, and recognise their continuing connection to country as Western Sydney continues to evolve and expand.

Western Sydney has emerged as one of the fastest growing locations for hyperscale data centres. We already supply about 20 data centres across our network, representing around 900 megawatts of installed capacity. If fully utilised, that is the typical demand we see on a network basis daily for 1.2 million homes. We are also progressing a significant pipeline of additional projects. Of the 15 projects fast-tracked through the New South Wales Investment Delivery Authority, 11 are within our network. Importantly, in New South Wales, electricity networks' revenues are capped and regulated by the Australian Energy Regulator. When large new loads connect

efficiently to the system, fixed network costs are spread across more units of electricity demand, which can place downward pressure on electricity bills for existing customers. But those benefits depend on coordinated planning. Poorer located connections can drive inefficient network augmentation, crowd out future capacity or reduce broader system value.

Our operational experience also highlights why more flexible connection frameworks are becoming increasingly important. In practice, data centre utilisation often ramps progressively over time rather than immediately reaching contracted capacity, which is broadly consistent with how hyperscale facilities are commissioned globally. What we are seeing, however, is newer AI applications appear to be accelerating that ramp-up rate. However, we're only just beginning to see those requests emerge locally. It reinforces for us the need for frameworks that balance timely access to capacity with efficient utilisation of shared infrastructure, while minimising the risks of cross-subsidisation, inefficient utilisation of allocated capacity, and the risk of stranded assets.

That is why networks such as Endeavour Energy are developing anticipatory concepts such as the Western Sydney Energy Hub. The hub aims to cluster data centre connections within a precinct, supported by shared enabling infrastructure, rather than potentially duplicating assets customer by customer. We are working with other utilities and industry to build on investment already underway, including the jointly funded bulk supply point and network works surrounding Kemps Creek. Importantly, these precinct-based models are not just about connecting large loads; they are also about creating opportunities to better integrate storage, renewable generation and flexible demand within shared precincts, improving infrastructure utilisation and enabling more coordinated long-term investment across the system.

We are increasingly incorporating these types of scenarios into our long-term distribution system planning, pushing networks to think beyond traditional project-by-project or customer-by-customer approaches and toward more adaptive and flexible operating and planning models. We believe this is the kind of strategic, precinct-based approach that should sit at the centre of New South Wales response to data centre growth. I'll now hand over to Annie Pearson from Essential Energy, who will speak to the opportunities in regional New South Wales.

ANNIE PEARSON: Thank you, Chair and Committee members, for the opportunity to appear here today. Essential Energy is a State-owned corporation and is the regional distribution network service provider for New South Wales. We manage more than 183,000 kilometres of power lines, covering 95 per cent of the State, serving around 900,000 connections across rural, regional and remote communities. Regional New South Wales and regional networks can play a bigger role in the State's future data centre strategy. Well-planned regional data centres can deliver customer, system and economic benefits. Essential Energy's experience with data centres is somewhat different from Endeavour and Ausgrid. Data centre development in our network area remains comparatively modest.

We are not currently experiencing the same concentration of data centre connection pressure. Policy responses, therefore, should reflect this and should not unintentionally raise barriers for efficient regional development. Importantly, more load or customer connections on the network help to lower the cost to serve and therefore can help to put downward pressure on network charges for our customers. Where resources and community support exist, regional data centres could ease pressure on constrained parts of the electricity system, make better use of existing assets, support co-location with new forms of generation and storage, and provide economic opportunities. But not every regional location will be suitable.

Data centre development depends on more than network capacity alone. It also depends on other utilities such as fibre, water, land use and local community support. Those factors need to be considered together, and considered early, before sites are locked in and avoidable network or planning constraints emerge. Government can improve regional investment readiness. Essential Energy is ready to help identify locations where data centre demand could be efficiently integrated with available network capacity, renewable generation and other requirements. We can do this by sharing network insights, hosting capacity information and connection expertise. We look forward to assisting the Committee and the New South Wales Government so that data centre growth is efficient, fair and aligned with the interests of all New South Wales energy consumers and regional communities. We would be very pleased to take any questions that you have.

The CHAIR: Thanks for your very comprehensive submission and for your opening statements. I already have a hundred questions, but I will start with a few and then pass over to my colleagues. Starting with that figure of between 17 to 30 per cent. As I understand it, at the moment data centres account for about 4 per cent of total operational load in New South Wales and the projection is that by 2040, data centres will account for between 17 to 30 per cent. Can you tell us what that means in terms of actual energy demand levels? How much energy is that exactly?

COLIN CRISAFULLI: One of the charts that sits within our submission talks to that growing to the area of about 20 terawatt hours—I believe that is the figure. For context, that is about the consumption that sits on Endeavour Energy's entire network today for that scale. That's largely a residential customer base as well as a manufacturing and industry base. But to put that into context, those sorts of figures would be the equivalent of adding, for instance, an Endeavour Energy back onto the grid, or about 10 per cent of the national figures.

The CHAIR: We were told in our hearing last week that data centres usually make multiple connection requests. They will ask to connect to the distribution level but then also directly to a transmission level. My understanding is that they will only end up needing one of those. Is there a world where they could take up both connection availabilities? Forgive my ignorance, but is there a case where that one might go down, so I need this one, so that they could have both?

FATIMA BAZZI: Because energy is a critical component for these data centres to get their DA approved and get the customer commitment, what tends to happen is they do multiple applications across the eastern seaboard. From there they see which one gets approved and then they will withdraw the other application. Sometimes they are looking for multiple connection points because, for example, they might want 500 megawatts, and therefore Ausgrid might be able to give them 200, Transgrid might be able to give them another 200 and Endeavour might be able to give them 100. There is definitely duplication in the system. That is absolutely why there is that 17 to 30 per cent figure, because we recognise that that is definitely an area of opportunity.

Endeavour, Essential and Transgrid are currently working together to look at our pipelines to understand where the duplication currently sits so we can really clearly see what the true demand is. It's the area that we really need to work on around forecasting and making sure that our forecasting methodology is consistent and that methodology then is fed into the AEMO forecast so we understand what the true demand is. Just note it's a really dynamic environment, it's very fast paced, so what I say today might be very different in six months based on demand requirements and just how much is coming onto the Australian seaboard.

The CHAIR: Again, I do wish I could just go off and do some sort of training in how to be a sparky. When we're talking about projecting demand, is there also an element—maybe this is wishful thinking—that they'll become more energy efficient? A data centre today at the same size in 10 years may not require as much energy because it has improved its efficiency. Is that also part of that range that you're thinking about?

FATIMA BAZZI: That's probably a question best placed for the data centres, but we have seen over the last few years that the technology continues to change. A personal theory of mine is, because there's so much demand, the technology will catch up, but it'll probably just catch up enough to capture the demand that's needed. But it's a very dynamic environment. The technology around the chips is really where that energy usage is coming from, so the chips inside the data centre, and that technology is changing year on year. So, absolutely, I think your theory is correct. How it eventuates, the timing of that, is yet to be seen.

ANNIE PEARSON: It's important to note that data centres, in many respects, are like any other consumer. They are constantly thinking about whether they need to become more efficient as their energy costs increase. So yes, they're large and they're new, but they are large consumers of energy and any large consumer, where energy is a huge input into their cost base, will do what they can to become more efficient. In that respect, they're probably no different from any other consumer.

The CHAIR: Is there any benefit in regulations saying that you shouldn't have multiple connections? If that's causing a concern in terms of being able to model accurately, is there any case to say, "Well, actually, can you just pick one?"

ANNIE PEARSON: Can I just take it generally first? There's modelling and then there's the reality that we experience as networks, and I think we always have to take into account both. Modelling is one input, but the reality on the ground is another, and that should inform our decision-making as well. What we see, and it is the same for generation as well as large load, like data centres, is they will often make multiple connections. For example, on our network, we probably have in excess of 21 gigawatts of renewable generation seeking to connect. Maybe 10 per cent of them will end up connecting. We know this. It causes us some problems, as Fatima has highlighted, but at the end of the day the most organised, the most developed projects get up. So, in a sense, the idea of some proponents blocking others, practically, does tend to sort itself out. It certainly could be more efficient, but that problem does seem to address itself.

COLIN CRISAFULLI: If I might add, I think we have found data centre applicants will look for multiple sites as well because they require multiple sites. In many situations they have, from what I understand, availability zones, so looking at locating multiple data centres within, for instance, Sydney to be able to provide that redundancy to their customers. So there's an element within there of it's part of their business model that they do that. But I think there is a reality that they do look for multiple connections to try and see which one is maybe

most cost effective, which one is faster because, depending on the location within that network, there are local situations that do drive that. That's why we've probably tried to highlight how do we get on the front foot with this, how do we identify those sites more at a precinct level to be able to encourage the connections within defined precincts rather than just a scattering across Sydney.

The Hon. SCOTT FARLOW: In terms of those precincts, do you do some form of mapping as to availability, and where are the areas where you have opportunities to be able to accelerate data centre development at the moment?

COLIN CRISAFULLI: We do at the moment. At the moment we do publish on an annual basis, hosting capacity across our network. Where we are looking at accelerating that is we've done that on the basis of where is there existing capacity. The more we've started to look at this, particularly with data centres, is looking at what is the next least cost increment in the network. So it is starting to look at different ways in which we're not necessarily just publishing hosting capacity, for instance, but actually what is the next least cost component of that hosting capacity as well.

FATIMA BAZZI: I might just add to that. One of our recommendations is coordinated hubs or precincts. The reason we want that is because, at the moment, the national electricity rules are stated in a way where any customer comes through any of our connections processes, and we have to proceed with that application and assess it. Therefore, they might pick a location that is already tapped out from a hosting capacity perspective but we still have to go through that process. Therefore, that's why our pipeline looks so big, because a lot of that pipeline is sitting in assessment phase to go, "Can this customer reasonably connect or not?" If we were moving into a space where New South Wales decided we're going to have these six hubs across New South Wales—that's where there's available capacity, fibre, water—then it really clears out a lot of the noise and that opportunity for these customers to go, "I want to be located in X location," where we know that that location isn't the right spot for those customers. At the moment the customers have the right to do that. We have an obligation to address that and we do so.

The Hon. SCOTT FARLOW: But you're wasting everyone's time and money, effectively, in this process.

FATIMA BAZZI: Potentially. It also allows us then to forward look what asset upgrades are required in the future. If we know, for example, that there are six key hubs in New South Wales, then we can foresee that five-, 10-, 15-year requirement around data centre need, EV charging, housing and all of those other factors that we consider in our planning process to then be able to go, "This is the upgrade," and actually plan for that versus, at the moment, we have to react according to what's in the pipeline.

The Hon. SCOTT FARLOW: While you have that electricity mapping, to your point, Ms Bazzi, what we've learnt in this inquiry is there's a combination of electricity and water. Do you have any interaction with Sydney Water or local water authorities when it comes to Essential Energy as to where that water availability might be?

FATIMA BAZZI: As part of our process, we've put in milestones as customers progress through the pipeline. We ask for things like, "Have you purchased the land? Do you have a DA approved? Has Sydney Water ticked off that you can actually access the water?", and so on and so forth. That is a process that we've put in place, and that has allowed us to remove a lot of people out of our queue. Do we work with Sydney Water? Yes, but it's outside of our jurisdiction to really go, "Can we approve or not approve a customer based on water requirements?"

The Hon. SCOTT FARLOW: You've got your remit.

FATIMA BAZZI: We do work with them. Colin does have some work underway in the Endeavour space that he might want to talk to around the hub.

COLIN CRISAFULLI: Yes. While there hasn't needed to be an obligation for a long time around integration of, for instance, energy and water, I think we're finding increasingly, yes, there is the congruence of the maps in which there is capacity, or call it that next marginal improvement area. We work very closely with Sydney Water on that. I would say it is early days because I think we're still learning essentially the long-term planning between the two parties and how it is we're going to tackle the challenge and opportunity that sits around data centres. It hasn't been a process in which we've had continual joint planning. It is done more at a project level rather than a system-wide level.

ANNIE PEARSON: We haven't done it with water utilities in regional New South Wales because we haven't had the same level of interest. However, we have coordinated with many local councils when it comes to identifying appropriate locations, say, for EV charging infrastructure. We have also undertaken a fairly extensive market-sounding exercise identifying areas across our network where large battery storage can connect. There's

precedent for working with other agencies there. We've just chosen to do that, but we could certainly do it in the context of data centres and local water authorities.

The Hon. MARK BUTTIGIEG: To follow up on the Chair's questioning about load, you said it was 5 per cent and then projected to be 30 per cent by 2040, was it?

FATIMA BAZZI: It's between 17 and 30 per cent.

The Hon. MARK BUTTIGIEG: Presumably, this is a fairly immature market in terms of your ability to predict stable growth because you are still learning the patterns. Given the multiplicity of applications, is it too early to predict a stable percentage of uptake, or are you starting to get a handle on it? How accurate is that projection, is my question.

COLIN CRISAFULLI: It's a great question, I think, and it is one we've been tackling as an industry for some time. We spent a fair bit of time in the development, and we gave reference to it, in the distribution system plan across the three New South Wales networks to try and put some more evidence to that. I think we have found very similar examples based on international experience as to what hyperscale of data centres and cloud compute looks like. We do see that as they come on, they do start to consume and then they take time as they progressively increase their demand as they fill out, presumably, their data halls for that. A lot of the forecasting and data that is within here is based on what we believe is actually quite a conservative forecast of what that demand is.

I think the part we are most challenged with at the moment, if we're being very honest, is the artificial intelligence and large language model demand. We haven't seen that significantly within our jurisdiction at this point. But I think that's where we're seeing probably the requests for that consumption coming in much faster and with a much faster rate of rise. That's probably the area, I think, we are uncertain about. But I think what we've placed, particularly within our forecasts, has been quite conservative based on what we're seeing in the pipeline and also where we're seeing the data centres invest pretty heavy capital into the enabling infrastructure—the connection infrastructure, the data centre itself. So it is real. I think it's just how fast and how far does it go.

The Hon. MARK BUTTIGIEG: So it's likely to be much bigger than that?

COLIN CRISAFULLI: It's potentially much bigger.

The Hon. MARK BUTTIGIEG: Does it factor in the point we made before about the technology getting better and better and requiring less cooling and that sort of thing? Or is that too nebulous to get a handle on?

COLIN CRISAFULLI: We've tested that, particularly within the connection requests that come in. What is the ultimate forecast and how do we potentially predict technology improvements in that time? I think it's safe to say I don't think we have a sound assumption sitting behind that, nor have we found probably necessarily a sound assumption that's consistent across data centre applicants as well. I think it's probably acknowledging that this is an uncertain area for everyone.

ANNIE PEARSON: It's uncertain because we have an electrification process also occurring on our networks as well as consumers gradually go off gas and other forms or other fuels on to electricity. So it's not happening in isolation. It's happening with more growth generally.

The Hon. JACQUI MUNRO: Your submission notes a significant disparity between the AEMO ISP projections and your own. Could you just talk us through what that disparity is? Can you explain that disparity? How should we be thinking about that? We often refer to AEMO, and it would be helpful to understand that.

FATIMA BAZZI: I can explain it in practical terms based on current pipeline so you can get a feel for how we determine these numbers. Currently, Ausgrid's pipeline for data centres is 7.5 gigawatts. Of that 7.5, we have 5.2 in planning assessment. So those connections haven't been approved. They're all of the ones that are sitting in the pipeline that we're assessing to say, "Can we reasonably connect these customers? Will these customers have the evidence through the lifecycle of this project to be able to continue to proceed?" Our forecast to AEMO is approximately 2.2 gigawatts.

Although the numbers feel big, when you break down the numbers, a lot of it sits in that front end of that application process. We recognise that the forecasting process needs to be more consistent across our organisations and we're working through that process. AEMO's figures would probably be our conservative number that we say, "Yes, we definitely think this is what's going to happen." What happens in the next 18 months will really be telling to determine whether that number is going to be bigger than what we've put forward would be kind of a broad statement.

The Hon. JACQUI MUNRO: I note that one of your recommendations is to have a more consistent modelling methodology. Where do you think that responsibility lies in terms of determining what that methodology looks like?

COLIN CRISAFULLI: I'm happy to take it first. I'd acknowledge, I think, AEMO's forecast and plan is built on a very robust engagement process, very robust econometric modelling. I think they openly acknowledge the challenge in forecasting data centres, and if you take the forecast they had two years ago to the forecast that's currently here, obviously there's already a significant change within that. I'd expect in the next cycle that again more information will come to light and that forecast will be updated to reflect the reality.

I think, as three New South Wales distribution networks, we've been more actively working with AEMO on the forecasting. To speak to what has probably been the more traditional approach, AEMO has traditionally relied upon its transmission network operators to support its forecasting methodology and that's because we weren't having these large centralised loads connect onto distribution networks. So, increasingly, all the networks and AEMO are working together to be able to refine that forecasting. But I think the open acknowledgement within this is there is a lot of uncertainty and all the parties are trying to grapple with that uncertainty in that forecast. But I think the process itself has been improving over time. I think the forecast will be updated to reflect what we're seeing on the ground, for instance. But I think it does still open the challenge of are we treating them consistently? How do we potentially compare them between States? How do we particularly compare them between networks? Because there are probably small differences between all of us.

ANNIE PEARSON: The AEMC is considering a rule change on this point right now, a rule change that came from Energy Consumers Australia, to look at how distribution networks planning can be more coordinated. The New South Wales Government is also considering the outcomes of its own transmission frameworks review, and the three of us are working with the New South Wales Government and with the AEMC to try to make sure that our respective planning, the many reports and the huge amount of work that we all do is coordinated and fits together. But where it is different—and it will be different at times—we're very clear on what those differences are, because there is no one truth here really and none of us has a crystal ball and is going to be right on this. We will all be wrong in one way or another. I think the most important thing with all of this is to be coordinated, to be clear about our assumptions and to be clear about how they all fit in alongside of each other.

The CHAIR: I was interested, when we talked about this causer-pays framework, in the mention of bank guarantees being required in some cases. Obviously there's an interest in data centre customers paying 100 per cent of their connection costs plus their share of network costs. But that issue of what happens if a data centre operator goes insolvent and is then not effectively paying their fair share—what proportion of data centre connections currently have that sort of guarantee of minimum revenue attached to them?

FATIMA BAZZI: Ausgrid has always put bank guarantees against all large load customers in a contractual sense. Over the last two to three years, we have further reinforced some of those contractual arrangements to ensure that in the event that those data centre customers decide, "We're done, we're out, we're leaving," they pay for the full cost of the augmentation and the cost of that infrastructure before they exit. Bank guarantees are legally enforceable. Obviously, data centres have ramp rates. They usually ramp up over a five-year period. What we are also doing is ensuring that those customers are paying for that full amount from the get-go. If they don't actually meet their ramp rate, they're still going to have to pay for that energy usage that they have committed to within the system. Bank guarantees are something that we've put in all our contracts. I don't know if that's consistent across all of New South Wales or in other States, but it is a mechanism that we have in place.

The CHAIR: Is that over the life of the asset?

FATIMA BAZZI: Correct. It's not just in the construction phase. It's also when they're up and running. Say in year five of them going live and they're fully energised, they make a decision—we have yet to see this—to exit the system, that bank guarantee is enforceable to ensure that any costs that still sits within the life of the asset can be legally received.

The CHAIR: Is that similar with the other DNSPs?

COLIN CRISAFULLI: Similar. I'd say for the majority of connected data centres on Endeavour Energy's network at the moment, the vast majority of the infrastructure was funded by the data centre applicants themselves, so there hasn't needed to be a direct guarantee, because there was no upstream work that would have potentially been funded by the existing customer base if the data centre was to disconnect or go insolvent. What we're finding is the use of guarantees becoming increasingly important as we start to move into this space where, actually, it does start to require more upstream network augmentation and reconfiguration, just to make sure that that augmentation doesn't necessarily get borne across the customer base if that data centre went insolvent.

The CHAIR: The other question that comes up a lot in this space is the New South Wales Electricity Infrastructure Roadmap charges, which don't apply to the transmission connected data centres. What's the approximate annual value that we're missing out on in road map charges that are currently being avoided by data centres who are directly connecting?

FATIMA BAZZI: I understand you have Minister Sharpe later on today. That might be a question best placed for the Minister. They haven't actually released the full road map costs. They kick off from FY27, so in a few months customers will start to see those road map costs. Today those road map costs will only be spread against distribution customers, so across our customer base. In the past, large load customers have not ever really connected to the transmission network. This is a new space for the transmission networks. Therefore we believe that we do need to review that scheme, because it isn't really equitable now for the customer base road map costs just to be borne on the distribution customers.

The CHAIR: That data, though, as to how much we're missing out on is held by government, not by—

FATIMA BAZZI: Correct.

The CHAIR: Understood.

FATIMA BAZZI: So government will provide us with visibility of what those costs will be, we'll apply that into the network portion of our customer bill and that will be pushed through to the retailer. That's generally the process.

The Hon. MARK BUTTIGIEG: I just want to explore this point that you raised in your submission and in your opening statements as well, I think. The network has a certain amount of capacity and it's built for that way to cater for very, very small windows of peak demand. And I think with the Ausgrid network—and I just remind the Committee that the Government fully owns Essential Energy and is a 49½ per cent shareholder in both Ausgrid and Essential Energy. I think for Ausgrid, Ms Bazzi, is it 10 gigawatts?

FATIMA BAZZI: The assets that are built provide capacity for 10 gigawatts of energy. We utilise that maximum demand between 5 and 5.5, and at minimum demand it's much lower than that. That's when we talk about the opportunity to reduce customer bills. I'll give you a broad example. If these data centres were to come onto the Ausgrid network and use up one additional gigawatt of capacity on our network, that should reduce all customer bills by approximately \$50. The best example—

The Hon. MARK BUTTIGIEG: Just so that people understand, this is because consumers are paying for that infrastructure—

FATIMA BAZZI: Consumers are paying for the 10 gigawatts, correct.

The Hon. MARK BUTTIGIEG: —to facilitate those air-conditioning days where everyone's hot and puts their—but if we get load connecting, then there's more money to go around and that cross-subsidises the residential bills, right?

FATIMA BAZZI: It's like if you were to catch an Uber to the airport, but you decided to share the Uber with the other Committee members, you are spreading the cost. It's exactly the same. Our customer base pays for the 10 gigawatts because they are the assets on our network. So the more utilisation we have on our network, the cost gets spread across.

ANNIE PEARSON: The revenues that we can earn are set by the regulator, so the more users, the lower the cost is to serve each customer.

COLIN CRISAFULLI: I think your point is very critical for Endeavour Energy's network. We do have a large industrial base, particularly down in the Illawarra. A typical demand on a typical day—not a sunny day but on a day like today—would be between 1.5 and two gigawatts of demand. On a very sunny day, that'll drop below one gigawatt because the solar systems are all generating and pushing their energy back into the system. On those peak hot days, it is about 4.2 gigawatts, driven largely by the residential volatility and air-conditioning load, particularly in Western Sydney.

The Hon. MARK BUTTIGIEG: Is there a point where, because all the data centres get connected—plus you've got your residential load and you have to account for peak load—that you then have to build more infrastructure? At the moment, there's probably too much excess capacity, right?

FATIMA BAZZI: Correct.

The Hon. MARK BUTTIGIEG: That's what's driving high bills. You've got to get those closer together. Is that what you're saying?

COLIN CRISAFULLI: It gets relatively complex. That leads a lot to the precinct-based thinking we're going through. How do you size it for the type of load that's going in there? Essentially, what we're seeing with data centres is a fairly flat profile—like 24/7 flat with a little bit of a surge in the middle of the day, but not much. That's very different to how we size, for instance, a suburb, which will be sized for those couple of days a year, but then for most of the year actually run at a fairly low utilisation. Sometimes it is around looking at the right

infrastructure in the right location and being able to actually co-locate types of customers, which can be beneficial because you get greater utilisation of the infrastructure base.

The Hon. MARK BUTTIGIEG: Ms Pearson, is that more accentuated for Essential because you've got this extensive network feeding the whole of regional New South Wales? You've got to have electricity lines, transformers and all the rest of it everywhere, but then your load, I imagine, would be fairly low compared to the capacity required. Notwithstanding the challenges you outlined with fibre and water, theoretically, if you put data centres out in the regions, that would help greatly, wouldn't it?

ANNIE PEARSON: It would help enormously because our network is underutilised. We also have, as you may be aware, a number of generator and storage proponents seeking to connect to our network. With some good planning, we may very well be able to identify areas where there's capacity for data centres and to have generation and storage close by as well. This would be very beneficial for our customers. We have, I think, four to five customers per kilometre of network but I think in Ausgrid's network, it is more like 45 customers. That's the difference that we're looking at. It's one of the reasons why in my opening comments I made the point that whatever policy response we take, it needs to be nuanced to reflect that this regional network is different and it has capacity. We want to make the most of any regional economic opportunities.

The Hon. SCOTT FARLOW: If I could just go back to that question in terms of building the network out for those four or five days—what's the impact of data centres on those days? I imagine there's some commensurate increase but from your discussion about consistency of data centre loads, typically, that variance may not be as much?

COLIN CRISAFULLI: It is obviously a much more predictable load once it's in operation. Being able to locate it and plan for it means that you can actually look at your infrastructure decisions to make sure you don't, for instance, assume it's going to behave like a residential customer. We take that into account in how we try to optimise the infrastructure required for that data centre. It's the same practice we apply for all customers. For instance, we have batteries often connecting adjacent data centres. You can assume that they're both consuming at the same time and that would actually result in overbuild, or you can actually use the battery, for instance, to reduce the demand that sits within that data centre or respond to the residential peak. So a lot of it is around how you coordinate the types of customers and the types of infrastructure to make sure that you don't have to build a big network to supply a demand that may not materialise.

The Hon. SCOTT FARLOW: Those four or five days in terms of peak air conditioners—I take it that, with data centres, because of the mix of cooling and there being a lot more water-cooling that goes into the operation, the demand on the systems might be more from the water network rather than from the electricity network. Is that a correct assumption or am I wrong there?

COLIN CRISAFULLI: I probably can't answer that one, but certainly what we've seen from the electricity perspective is not a significant amount of shift in that profile.

FATIMA BAZZI: We had an example recently where we had one of our peak days and what we did actually ask the data centre to do to kind of rectify was to actually take their load down just for a few hours to allow the market to stabilise and then they went back up. If we're really smart and if we really plan well, what we can do is we can actually utilise these data centre customers to peak in and out as required, where they can, because obviously with some data centres you don't want them to be flexible because of the type of customer they're servicing at the end, while there are other data centres, which makes sense, where they can curtail their demand a little bit for a few hours just to even out the system for those peak periods.

The Hon. SCOTT FARLOW: How do you know the profile of the data centre's customers? Is that something that's shared with you or is it just a request you put out and it's a price mechanism effectively and they'll respond to the price mechanism?

FATIMA BAZZI: There's the process where they come in and connect and they give us a ramp rate, so we assume and we go, "Okay, this is how we're going to model it", and then there's the practicality. When they operationalise, we can see their demand, so based on that we can then start to really forecast that operational demand. They're different processes. The first one is a planning process. The second one is how we operationally manage our networks.

COLIN CRISAFULLI: A lot of our process improvements have been around validating a lot of those inputs, obviously based on experience and trying to understand not all data centres are the same. They all have very different business operating models. But I think we've also found, even in that up-front engagement, you can start to discuss things like flexibility, so flexible demand, because that would be a fantastic resource for the energy system if on those couple of days for a couple of hours they could reduce the level of demand. It can also accelerate their connection, but it can also support what I call a skinnier grid to be able to service that connection and push

downward pressure on bills. I think what's occurring at the moment is a more dynamic way in which we're thinking about these customers, and it's an opportunity that's only arisen just given the size and scale of them to be able to negotiate that and try to see what is the best path in the long-term interest of all consumers.

The Hon. SCOTT FARLOW: As a point of clarification or expansion in your submission, I'll quote the second last paragraph on page 5:

The total network price for distribution connected customers includes several components: distribution revenue recovery, metering charges, transmission revenue recovery and jurisdictional cost pass-through. Jurisdictional scheme costs (such as those associated with the NSW Electricity Infrastructure Roadmap) are expected to increase over time and are correlated with customer consumption. Currently, if data centres connect at the transmission network level they would not pay their contribution towards the NSW Electricity Infrastructure Roadmap.

I'm just interested in that point and how that works and how potentially we could ensure that they did pay if that was the aim of government.

FATIMA BAZZI: That is a policy decision by the New South Wales Government, so it would be just a policy change and then that cost would be spread across both transmission and distribution customers. It's kind of outside of our control. We're the catch of it. Therefore, I think it's a discussion with New South Wales government on how they're going to adjust that policy.

ANNIE PEARSON: We have been encouraging Government—the three of us over some time—to spread it across both distribution and transmission for that equity reason. There's a similar issue with battery storage. If battery storage connects to the transmission level, it doesn't attract those charges; if battery storage connects at the distribution level, it does. There are a number of those sorts of differences that we would like to remove as distribution networks and transmission networks become a little bit more similar in how they look and feel as the energy transition progresses.

The Hon. SCOTT FARLOW: To that point, is that part of why there's the double application, so to speak, that we were discussing earlier? No. I'll note the shaking of the head for Hansard.

ANNIE PEARSON: It may be a contributing factor. Large loads and large generation—they like to have connections at the transmission level and at the distribution level. They like to diversify because it's in their best interests to have a diversity of locations and connections.

COLIN CRISAFULLI: I reflect that the mechanism in which that cost recovery occurs is one that's been built over many, many years, including, for instance, NSW Climate Change Fund. That also funds a lot of schemes like solar for low-income homes and other schemes that come from that. It's been levied on distributors because that's where the customers were connected for a very long time. What's changed now is customers are connecting at both transmission and distribution, and so the time is now to be able to address a potential inequity. It will result in some fairly substantial differences in pricing for a connection to a transmission network, to a distribution network. Our challenge is: Is that by design or is that actually just an arbitrage that may not be what was intended?

The Hon. JACQUI MUNRO: May I go to some evidence that was provided to us last week. I don't know if you were watching, but the Secretary of the Premier's Department said that there would be no data centre strategy following the consultation that the Government is undertaking currently. Is that concerning to you?

FATIMA BAZZI: Whether it's a policy decision or a strategy, I do believe in Victoria they've done it quite well, as a starting point. They have managed to accelerate their approvals in a four-month period. I have customers who've been waiting 18 months to determine are they going to proceed or not proceed. It's not a great look for New South Wales and so, therefore, they are just moving to Victoria to connect. I believe it's in New South Wales' interest for us to open our doors to data centres and do it in a coordinated, managed way where we do mitigate the risks.

We can put forward papers, we can put forward recommendations, but what I think we all need to do is we need to understand that this is a dynamic environment. We should have a process that we adjust as the environment changes, and to enable the networks and these customers to move through that process, versus spending 18 months trying to make decisions. We need a coordination body that has decision-making rights. At the moment, everyone has a decision-making right for a part of the puzzle, and that makes it really, really tricky for customers coming into our environment and for all of us to collectively drive these outcomes.

ANNIE PEARSON: I think we will work with what we have.

The Hon. JACQUI MUNRO: You have to, really. You've got no choice.

ANNIE PEARSON: We have to. We work with what we have. Government has an EV charging infrastructure strategy but, on a practical level, Essential Energy has maps. It's worked with local providers, local councils. We do what we need to do to coordinate this activity because, in our case, we want the consumers in our

network—particularly in rural, regional and remote New South Wales—to get the same benefits as customers in metropolitan areas. That's what drives us. If we have Government policy to coordinate it, that's great—that's ideal. But the three of us all have a long history in working with what we have, working with local councils and other key community stakeholders to make things happen.

COLIN CRISAFULLI: I think it's work with what we have and then, obviously, we advocate to drive change so that we can get consistency and support a better outcome.

ANNIE PEARSON: We also have a rule change process. One of the advantages of the energy sector is that we have an independent rule maker who has absolutely no discretion not to consider a problem that we put to it, so we can actually achieve a lot of reform that way. It's not perfect and there are limitations on what they can do, but we can do a fair bit that way.

The Hon. JACQUI MUNRO: Your submission states:

The unprecedented, system-shaping scale and pace of data centre growth in some parts of the network cannot be managed through business-as-usual planning.

It specifically says that there should be more strategic planning of data centre locations. Your opening statement referred to a proactive strategy. Has the Government asked you for your input into whatever they're doing at this point? Are they proactively engaging with you?

FATIMA BAZZI: We have actively been working with the IDA, Investment NSW; we've been working with the energy department and the planning department. I believe what we need to do is just decide on the path forward so we can all support these customers through this in a way that benefits the broader population of New South Wales. There's definitely benefits. I just draw the Committee's attention to the Victorians. The Victorians just got their determination and Victorian households are forecast to save \$175 a year by FY30. That is amazing. That is really, really hard to do. A lot of that comes from them building data centres on their patch to drive that utilisation up and to drive the cost down. I draw the attention of the inquiry to that because I want that for New South Wales. I want that for our customers. Road map costs are going to be high. At a minimum, if we can find a way to break even, that would be an amazing outcome for New South Wales.

The Hon. MARK BUTTIGIEG: Can I ask you a question about where this whole thing is going in terms of renewables? As more solar and wind and battery storage come onto the network, the distribution networks that you run become more critical because you have to wield the power around and share it. Are you facing regulatory hurdles in terms of your ability to leverage that infrastructure into that transition with community batteries and the like? Sorry, this is a bit long-winded. When data centres are going in, has there been any thought to charging the data centres for things like community batteries that you're putting in to soak up that load?

COLIN CRISAFULLI: I'm happy to go first. It's probably safe to say it's mixed. Endeavour Energy has invested heavily, predominantly in digital infrastructure to be able to connect and develop new value propositions for customers. For instance, we launched a flexible exports product. The idea is customers can export more solar most of the time because it's interacting with the system and making sure the system stays secure. But most of the time, they get a lot more solar out of the system. A lot of our investment, and I think it's across many of the networks, has been in digital infrastructure—things like customer management systems, customer consent management systems, digital infrastructure to monitor the network and identify if there are issues and constraints. There's been a lot of support, admittedly, for that type of investment which helps avoid costly investment because it's no longer set and forget.

You don't just put out, for instance, a transformer and come back in 50 years and replace it. It's actually knowing when things are operating at their limit. It is to drive greater utilisation of the grid. Where we do find challenge is where we try to come up with fairly new models, novel models—things like community batteries, things like EV-charging infrastructure on poles or looking at local energy precincts to be able to generate as well as sell energy within that precinct. These are areas that are probably novel, so the regulatory frameworks haven't kept pace. So we're not necessarily seeing support for that, but we're also acknowledging we're all pushing into this area to find new ways in which we solve the energy transition. I'd say it's a mixed result, and it's probably just the path we're going down, being able to advocate for change.

ANNIE PEARSON: I would agree with Colin. To the extent that we need to make investments to carry out our standard BAU distribution service role, yes, we can do that. The AER holds us to a high standard, as it should. We can make those investments, which we've had to, because our network is being used in a different way. I think where it comes to newer, novel-type services, it's a little trickier. It's particularly tricky going back to the regional-metropolitan divide as well. I think, from Essential Energy's perspective, our customers aren't seeing the benefits of a lot of the energy transition, such as access to EV charging infrastructure, yet the regulators are less likely or less comfortable with treating us differently and allowing us to provide services that perhaps wouldn't

need to be provided in other areas. I agree with Colin: When it's the standard distribution service function, we can do what we need to do; it's those other newer services that our customers want where it's a little trickier.

The Hon. JACQUI MUNRO: Just to clarify, you think that there is a real opportunity for regional areas to be home for certain types of data centres to assist with grid stability, reliability and to reduce prices?

ANNIE PEARSON: Absolutely and definitely. We want regional and rural customers to benefit from these technological changes and one way is through a stronger network that comes with bigger load and also through more efficient charging or more efficient utilisation, I think, which leads to lower charges.

The CHAIR: We heard evidence last week from Data Centres Australia—or maybe that was the week before, but definitely last week from the IDA—talking about quite a long ramping profile. But in your submission you say that requested loads are now ramping to full capacity in one to two years rather than a longer period of time. Can you explain the disparity?

COLIN CRISAFULLI: We found that, typically, most of the data centre connections of the 20 we've got at the moment tend to switch on at about 20 per cent and then they ramp up over about 10 years to a level of utilisation. What we're probably alluding to in the statement is that we're now receiving requests that are suggesting a ramp rate of two years. We haven't seen those yet materialise in any of our actual operational experience, but I think the demand has increased and we've linked that largely to different opportunities that obviously these data centre proponents are starting to identify. But, by and large, our profile that we've seen to date has been—call it that five- to 10-year ramp rate. That's quite manageable from a network and from an infrastructure perspective. I think if it was genuinely to ramp up at one to two years, we would be challenged with that within our current frameworks.

The CHAIR: That does, unfortunately, bring us to time. Thank you so much for your appearance. To the extent there were questions taken on notice—and I know that there will be supplementary questions—the secretariat will be in touch.

(The witnesses withdrew.)

Mr MARTIN KENNEDY, General Manager Market Operations and Grid, Clean Energy Council, affirmed and examined

The CHAIR: Good morning. We now welcome our next witness. Would you like to commence with a short opening statement?

MARTIN KENNEDY: I would.

The CHAIR: Please go ahead.

MARTIN KENNEDY: Chair and Committee members, thank you for the opportunity to appear today. Data centres have become a shorthand for artificial intelligence, but most of the digital infrastructure Australians rely on every day has nothing to do with ChatGPT, AI or generated cat videos. These facilities in fact carry our banking systems, emergency services communications, CCTV networks, hospital records, university data, defence infrastructure and all the other cloud services that businesses and individuals rely on every hour of every day, every Teams meeting, every government database, every online transaction. This needs to be viewed as essential digital infrastructure for the future.

We build our dams to store Sydney's drinking water in or near Sydney. We would not choose to keep it in Texas or Shenzhen. Data is moving into the same category. There is a clear national interest in storing and processing as much of it as we can on Australian soil, under Australian law and powered by Australian energy. One of the biggest opportunities data centres present is their ability to help unlock renewable energy projects stuck in the investment pipeline currently. AEMO, the market operator, expects data centre demand to grow roughly fivefold by 2035, reaching around 9 per cent of national electricity market consumption. In New South Wales specifically, this could exceed 15 per cent.

This is an enormous new source of electricity demand entering the system. From a market and grid perspective, that matters, because data centres are exactly the kind of customer that many renewable energy projects need. They are large-scale, long-duration users of electricity with strong incentives to sign long-term power purchase agreements with new wind, solar and storage projects. Those contracts provide the revenue certainty developers need to reach financial close, which at the moment is a major issue for Australian clean energy. Globally, players such as Amazon, Microsoft, Google and Meta are the four largest corporate buyers of renewable energy on Earth, with over 90 gigawatts of renewable PPAs already contracted internationally. In Australia, Amazon alone has signed PPAs with around eight local renewable projects, totalling over a gigawatt. This shows you the market to power data centres with renewables already exists.

Across New South Wales today there are projects sitting close to financial investment decision but struggling to get over the line. Large data centre customers and the load that they would bring and the contracts they would sign can help change that. Their demand can bring forward new generation, accelerate storage investment and help move projects out of the queue into construction and operation. This is not about clean electricity and data centres as a binary struggle. It is about a provider and a customer relationship that is mutually beneficial.

The third point is that Australia can absolutely power this industry through firmed renewable energy. The system can support it. A modern electricity system needs three things working together: generation, including wind and solar at scale, as well as rooftop and consumer resources; storage in the form of batteries and other long-duration assets, and firmed by gas; and transmission to move the energy from where it is generated to where the load sits. Behind-the-meter batteries at data centres can also help reduce peak stress and replace diesel backup in a cleaner, more sustainable manner.

Research commissioned by the Clean Energy Finance Corporation found that if new data centre load is not matched by new renewables supply, New South Wales wholesale prices in 2035 would be around 26 per cent higher, and emissions around 14 per cent higher. Matching new demand with new supply protects households, provides investment certainty and protects emissions at the same time. Australia has the resources, the workforce and the market design to build sovereign digital infrastructure powered by Australian wind, sun and storage, which is what we should be aiming for, in our view. Thank you. I'm happy to take questions.

The CHAIR: Your submission argues that data centre power purchase agreements can "underwrite the financial viability of new wind, solar and battery storage projects" and also add new net renewable capacity to the NEM. But the Climate Council's submission to this inquiry noted that in recent years there has been a decline in corporate PPAs underwriting genuinely new projects, with most attached to existing assets or underwritten by State-owned utilities. The Business Renewables Centre Australia's state of the Australian PPA market report raised similar concerns about additionality. What mechanism do you propose to ensure that data centre PPAs

genuinely support additional generation, rather than simply providing a revenue stream for projects that would have been built anyway?

MARTIN KENNEDY: I think this is a key point and it's one that we have put a fair amount of thought into. It's fair to say we're still refining our position as an organisation and an industry. Data centres, particularly hyperscale ones, are material amounts of new load, not dissimilar to a smelter. You wouldn't bring something like that into the system without also considering investment in the supply infrastructure to meet that new demand. Failure to do so equals adding new demand without new supply, which is only going to push prices in one direction. We feel there's a strong case to match the new demand with new supply. Firmed renewables are consistently being found to be the cheapest form of new-build supply in Australia.

Now the challenge that needs to be unlocked, in our view, is that at the moment the time to build a data centre is shorter than the time to build a renewable energy project, typically. This is where we're seeing players contracting with existing assets, signing PPAs with them rather than new build. Our view is that there's a natural opportunity here to support existing assets with a market exposure that they want to hedge in the near term. But it's also important to have additionality in the longer term; otherwise we get into that scenario of adding demand without adding complementary supply. In terms of how we work through an implementable mechanism there, our approach is a collaborative one, working with the data centre industry and our renewable members to figure out how we achieve those twin objectives.

The CHAIR: Would you support a requirement that data centre PPAs include an additionality clause—so something that actually requires that the PPA support a project that would not otherwise have proceeded?

MARTIN KENNEDY: That's one we would want to work through with them, maybe in the form of a working group or something, to collaboratively figure out what's going to work best. The evolution that we've seen of some of the advocacy in this space, from initially being focused on pure additionality of pure renewables to an acknowledgement that some degree of firming is also required, and an acknowledgement that if you're pulling the trigger on your renewable project and your data centre project at the same time, the data centre project is complete before the renewables one, particularly in the case of wind—there is some nuance to be worked through there in terms of how that gets brought into effect, but we see a natural win-win opportunity here for the data centre industry and the renewable industry to work together.

The CHAIR: Obviously these PPAs are directly negotiated between the data centres and the renewables project. These data centre companies consume a lot of energy and they're very motivated to minimise the cost to themselves. What confidence do you have that the prices being offered to renewable projects from these data centre companies are sufficient to bring those projects to a financial close?

MARTIN KENNEDY: I would say that both parties have a strong incentive to reach a mutually beneficial position. The data centres ultimately need to power their operations, energy being the biggest single cost in most instances and firmed renewables being consistently found—for instance, in CSIRO's GenCost report—to be the cheapest way to do that.

The CHAIR: We're hearing from quite a lot of renewable energy proponents—in particular, wind farms—saying that they don't feel that the PPA prices that they're being offered are sufficient, basically, in order to then reach that financial close. They feel that they're getting squeezed. Is that something that you're hearing as well?

MARTIN KENNEDY: There are a couple of challenges that I think are underpinning that. One is the currently very low level of large generation certificate prices, which doesn't really provide much of an investment signal for new-build renewables but does, in theory, enable electricity purchased with those certificates attached to be positioned as green. So I think there's a challenge with that pathway allowing people to make certain claims that aren't really driving genuine additionality.

The other related challenge is that there are projects that would have contracted PPAs in the early days of the RET that would be rolling off those into market exposures that they don't want to carry, and they would be looking to recontract PPAs effectively in competition with the new build projects. If you're taking a purist investment view, you've got an existing asset that is available with a known output profile, no construction or development risk, potentially outcompeting the new build assets that have at least some degree of risk in those areas. I think that's a challenge that needs to be worked through because those that have invested in the capacity previously have a right to earn a return on that. But, equally, we want to make sure that that's not at the expense of investment in the new build capacity we need to put in place.

The Hon. JACQUI MUNRO: Thanks for coming and for your submission. I actually referenced the energy newsletter last night in two of my contributions to Parliament and I was curious about basically the social licence for projects. We just heard that there is a big appetite and potential for data centres to roll out in regional

areas to lower prices and stabilise the grid. Are you seeing in your work that social licence for renewable rollout and data centre rollout as having similar pitfalls or risks in regional areas?

MARTIN KENNEDY: I would say there's a similarity in that, when it's done well, the assets are received favourably and, when it's done poorly, a lot of people get upset, which is understandable. We've seen plenty of examples of renewable projects that have contributed materially to community benefit funds and provided drought-proof revenue streams to farmers hosting the infrastructure. But then there are also examples where consultation processes maybe weren't as thorough as they could have been. In that sense, there's a similarity in that there are examples both in the positive and the negative.

I think one of the challenges with data centre siting is you need a few different things to come together, one of which is availability of favourable grid conditions, which is what you're talking to, and the energy needed to power the operations. But they also fundamentally need a high-speed, high-capacity fibre internet connection. I guess that's the bit that needs to be considered, in our view, in relation to these decisions around regional siting because, if we're just trying to solve for the best spot in the grid to put new load but there's no way for them to get the data in and out, we're boxing ourselves in.

The Hon. JACQUI MUNRO: It seems like there's a slowing pace of renewable energy rollouts in New South Wales at the moment. Could you speak to what needs to be done to improve that rollout so that we can have some more confidence that data centres can be a useful part of the mix?

MARTIN KENNEDY: This is something that, perhaps unsurprisingly, we spend a lot of time on. Last year I think there was about 4.4 gigawatts of capacity across the NEM that commissioned. If I refer to AEMO's Electricity Statement of Opportunities report last year, we need to be doing closer to 10 per annum on average. So there's a big uplift that needs to happen across the NEM to get us where we need to go. That's a challenge, but it's, in our view, certainly not insurmountable, and I would draw attention to the fact that it took us 10 years to get from 10 per cent renewable energy to 20 per cent, but it took us only five years to get from 20 to 40. As at Q4 last year we actually went slightly beyond 50 per cent renewable energy in the grid. The pace can accelerate; the capability is there. The challenge that we're seeing at the moment isn't so much at the construction and commissioning point. It's certainly not at the development stage.

We've got 350 gigawatts of development capacity sitting in AEMO's queue at the moment, which is five times more than the NEM currently has in total, way more than we need. So the projects are there and the capabilities there. What's missing, in our view, primarily, is the bankable investment signal that gets people to FID and gets construction underway. In New South Wales the primary program to try and address that would be the LTESA structure within the New South Wales road map. There was a new tender round recently launched to that effect, with a strong focus on getting contracts signed and projects commenced this year, which we view, obviously, as a very favourable development. We'll continue to work with them on whatever we can do to help accelerate, but I guess the headline message would be: Yes, there's a lot to do. We believe it's achievable and we believe it's in everyone's interest to achieve it. We're here to collaborate and help make that happen.

The Hon. JACQUI MUNRO: With the LTESA process, have you witnessed a delay in the most recent round coming out? There's been a gap between the rounds. How do we ensure that that is actually a mechanism that's utilised effectively?

MARTIN KENNEDY: It's a key point. I think there was a little bit of interplay between the capacity investment scheme process and the LTESA process. That to some degree explains the delay. I was looking at data recently—yesterday, in fact—that was suggesting that about 50 per cent of the LTESA awarded projects had reached FID. Obviously, there's 50 per cent in the queue that's been awarded that still needs to get there, and then there's the other capacity that still needs to be contracted in the remaining auctions. In terms of what needs to be done to accelerate those processes, that's probably a conversation we'd want to have directly with ASL and EnergyCo to work that through.

The Hon. JACQUI MUNRO: Finally, can you just talk me through that capacity investment scheme interaction, please?

MARTIN KENNEDY: The capacity investment scheme, projects in New South Wales were able to bid into that scheme, so there were New South Wales based renewable projects contracted through it. That was happening during the lull that you mentioned in the LTESA program.

The Hon. JACQUI MUNRO: How does that make it more difficult for investment to flow through? You seemed to indicate earlier that there was a problem between those programs interacting or—

MARTIN KENNEDY: No, sorry, that was not my intention. I was inferring that perhaps—and I don't know as a point of fact, but perhaps part of the reason for the apparent delay in LTESA rounds was because there were six rounds happening in that period that included projects in New South Wales.

The Hon. JACQUI MUNRO: Right, so there was a decision made to prioritise that, potentially.

MARTIN KENNEDY: Possibly. This is an inference.

The Hon. JACQUI MUNRO: Understood. Thank you.

The Hon. MARK BUTTIGIEG: I have a general question related to the previous evidence we heard and this dovetailing into the transition to renewables. What we heard on evidence from the three distributors who cover New South Wales is that because of all this excess capacity in the network, there's a big potential for data centres to connect and soak up some of those infrastructure costs and therefore subsidise electricity bills. If there's then some sort of a requirement for those data centres to be supplied with renewables, can you see a positive kind of virtuous circle there where consumers feel as though they are actually seeing a material reduction in bills because of the data centre load but also that is encouraging renewables to come online as well? Is that a fair assessment?

MARTIN KENNEDY: Yes, that's a good example of the win-win that can be possible here when we think about it in that way. The data that I have seen suggested that most data centres were looking to connect into that distribution system which, as we heard, does have some spare capacity in it. Obviously for consumers, it's a good outcome if that capacity cost is spread over some of this new load that's coming into the system.

The Hon. MARK BUTTIGIEG: Then the political appetite for the transition to renewables becomes greater because people are actually seeing material reductions in their bills, right?

MARTIN KENNEDY: Yes, absolutely. I would also mention on that point the latest default market offer determinations, which generally were lower across the NEM—and a big driver of that was lower wholesale prices, which are correlated with higher renewables in the wholesale market.

The CHAIR: Greenpeace released a report last week that stated:

... the frenzied rollout of AI data centres in Australia is set to derail the renewable energy transition, entrench gas and turbocharge climate pollution ...

They call for a moratorium on data centre builds. Why is the Clean Energy Council so at odds with one of the leading climate change advocacy organisations?

MARTIN KENNEDY: I think their position is probably accurate if you don't build the new-build renewables to support the data centres. The research that the Clean Energy Finance Corporation commissioned from Baringa shows that if you don't build the new-build renewables, it does push up prices, it does create reliability challenges, and it does threaten emissions targets.

The CHAIR: Your position then is reliant on the idea that there will be genuinely additional renewables coming onto the—

MARTIN KENNEDY: We view it as a win-win. You have renewable developers who are seeking long-term offtake, and you have data centre developers who are seeking to contract long-term, cost-effective electricity. We see that as a natural win-win.

The CHAIR: You then have the data centre in the Southern Highlands that wants to have a gas-powered plant in addition, so I guess that's also something that perhaps they've got in their minds?

MARTIN KENNEDY: I probably can't comment on the individual decisions of individual sites, but that obviously sits outside what we're suggesting.

The CHAIR: In your opening statement you said that most data centres aren't about ChatGPT et cetera but the critical government data. We've got evidence to this Committee from researchers and the transmission operators that approximately 84 per cent of new data centre load is for AI.

MARTIN KENNEDY: That's interesting.

The CHAIR: Also there's huge secrecy around what these data centres are being used for anyway. How do you make that statement?

MARTIN KENNEDY: I was basing my statement on the analysis that AEMO commissioned from Oxford Economics. That indicated that most of it was cloud computing and that AI was a material component, but a relatively smaller component than the data storage, backup and compute within the cloud computing.

The CHAIR: Which is at odds with the information we heard last week, in particular.

MARTIN KENNEDY: It's an issue of timing. I mean, maybe the Oxford Economics statement is as of now looking backwards and other statements seem to be more forward-looking about their pipeline, which I don't have visibility over.

The Hon. MARK BUTTIGIEG: Just to clarify that, was that study based on projected growth of data centres based on historical trends—in other words, banking, cloud computing and all the rest—without necessarily understanding the uptick in demand for AI? Or did they explicitly say, "We think that, because of the growth of AI demand, this is going to be the projected demand?"

MARTIN KENNEDY: I have to say I didn't look into that element of their methodology. The Oxford Economics report is publicly available, though, so you'd be able to dig into that detail.

The CHAIR: I think that's a good point. That probably explains it.

The Hon. SCOTT FARLOW: Mr Kennedy, whether it's AI or whether it's data centres for critical storage, AI is coming, if not already here, and it is growing. The question for this Committee and for the New South Wales Government comes down to whether it's worth having here, in New South Wales. Just looking in terms of your submission, you see that there are advantages for New South Wales in having data centres and data centre development in this State. Is that correct?

MARTIN KENNEDY: Yes.

The Hon. SCOTT FARLOW: Those advantages effectively come to supporting the renewable energy rollout?

MARTIN KENNEDY: Yes, among other things. Obviously, that's our primary focus as the Clean Energy Council: the jobs and the investment in regional communities where the renewable projects to power the data centres would be based.

The Hon. SCOTT FARLOW: If there was an alternative, if New South Wales decided to thumb its nose at data centre development in this State, what would you see, at least in your patch from the Clean Energy Council, as being the alternatives and where these data centres would go or what sort of arrangements they would have governing them?

MARTIN KENNEDY: My understanding from the evidence that I heard from the folks that were in here immediately before me was that it would most likely head to Victoria. That's what they seemed to be saying.

The Hon. SCOTT FARLOW: What are the arrangements in Victoria compared with New South Wales from a clean energy perspective? Are you aware of that?

MARTIN KENNEDY: In terms of the specific policy drivers for Victoria versus New South Wales, I would probably defer to the folks that gave testimony immediately before me. They seem to have some information on that.

The Hon. SCOTT FARLOW: But in terms of the requirements around clean energy, and your submission talks about having, I think, a recommendation that you ensure that there is a clean energy requirement for any data centre, is that something that exists in Victoria at present?

MARTIN KENNEDY: I would need to take that on notice.

The Hon. SCOTT FARLOW: If you could, that would be good.

The CHAIR: Thank you very much for making yourself available and for giving us the benefit of your insights and experience. To the extent that there were questions taken on notice, or more likely supplementary questions, the Committee secretariat will be in touch. That does conclude our time with you.

MARTIN KENNEDY: Thank you all.

(The witness withdrew.)

(Short adjournment)

Ms WENDY BLACK, Executive Director Policy, Business Council of Australia, sworn and examined

The CHAIR: We now welcome our next witness. Would you like to commence with a short opening statement?

WENDY BLACK: Yes, if I could. Thank you, Chair, and thank you to the Committee for the opportunity to appear today. The Business Council of Australia represents many of the nation's largest employers and investors across the economy, including companies investing in digital infrastructure, energy systems, advanced manufacturing, telecommunications and AI-enabled services. Importantly, whether directly or indirectly, all our members are reliant on data centres to run their businesses. That goes for businesses large and small, including all the businesses within the supply chains of our members.

Almost every Australian already relies on data centres, often without realising it, from accessing banking services, healthcare systems, government services or communications platforms. From a business and economic perspective, data centres are foundational economic infrastructure essential to a modern economy and society. They underpin cloud computing, cybersecurity, digital government services, financial systems, health systems, AI capability and the broader digital transformation of the economy. Importantly, they are also becoming central to Australia's future productivity performance. Every major industry is becoming more data intensive and more digitally enabled. Data centres provide the cloud infrastructure that businesses, governments and essential services rely on to store and process information. They are also supporting AI adoption across sectors such as health care, logistics, resources, education and financial services, which will increasingly depend on these for secure, reliable and scalable compute capacity.

That means the question before this Committee is bigger than individual facilities or planning approvals. It is whether New South Wales intends to remain globally competitive in a rapidly evolving digital economy. Does New South Wales want a seat at the table in the development of this technology and into the future? Countries across the region are competing aggressively for digital infrastructure investment. Australia starts from a position of strength, with strong institutions, renewable energy potential, high levels of digital adoption and a stable investment environment. These are advantages which we can further leverage to attract investment and jobs.

Investment in this sector is highly sensitive to policy certainty, planning efficiency, energy availability and regulatory consistency. If settings become fragmented, uncertain or excessively slow, investment will move elsewhere. That matters not only for economic growth, but increasingly for sovereign capability and national resilience. At the same time, communities rightly expect governments and industry to manage energy demand responsibly, maintain reliability and affordability in the electricity system, minimise environmental impacts and ensure infrastructure development occurs with proper planning and transparency. The answer is to acknowledge that we have sound policies in place which, if applied and coordinated, will enable New South Wales and Australia to gain the benefits of this investment.

From the Business Council's perspective, four priorities are critical. First, governments should recognise data centres as enabling economic infrastructure and as a strategic investment opportunity for New South Wales and Australia. Second, New South Wales and Commonwealth policy settings should remain aligned; a fragmented or inconsistent national framework risks undermining investment confidence and Australia's competitiveness. Third, planning and approvals systems must become faster, more predictable and more coordinated. Long approval time frames and inconsistent processes increase costs, delay investment and weaken Australia's position relative to competing jurisdictions. And fourth, infrastructure coordination matters; delivering digital infrastructure at scale requires closer coordination across energy, transmission, water, telecommunications, planning and workforce systems.

Importantly, the sector can support broader economic and energy objectives. Data centres are already underpinning investment in renewable energy storage and grid infrastructure through long-term energy purchasing agreements and infrastructure investment. They are also supporting high-value construction, engineering, digital and technical jobs across the economy. Australia has an opportunity to position itself as a trusted regional hub for the digital infrastructure and AI capability. The Business Council believes Australia can absolutely support both economic growth and responsible infrastructure development, but doing so requires governments to approach digital infrastructure strategically and cohesively, with an attitude to find solutions, not put up barriers. If we get this right, New South Wales can strengthen productivity growth, support higher living standards, attract investment, build sovereign capability and create long-term economic value while maintaining community confidence and supporting the energy transition. Thank you. I look forward to the questions from the Committee.

The Hon. SCOTT FARLOW: Thank you very much for your submission. In terms of the benefits you've outlined for business and its utilisation of data centres, does it matter if they're here or if they're overseas? What's the advantage to Australia and particularly New South Wales hosting them?

WENDY BLACK: I think there are the two main advantages. First is the capital expenditure that would come here. That is that broader economic growth piece. With the figures that came out yesterday for capital expenditure in the country, really it is data centres that are giving us that investment at the moment. It is a broad investment in construction, in the technology, creating these very skilled jobs, so there is a broader economic value. Then there is that wider role that they play that is the digital infrastructure that we need for businesses going forward. We are changing the way that our computer technology works. We're now having cloud technology. One of the positives of that is it is more energy efficient than having servers in every location, so that's a good one. But also when you look at the way that more digital services are being provided, it's much more efficient and much more productive that we're digital rather than paper.

The other part is the AI evolution that we're having at the moment. There is no doubt that this new technology is going to change the way that we do business. There are enormous productivity gains to be had from that. For those sorts of services to be available, it is much better that those data centres providing that compute power are located close. I'm not saying that all data centres are the same. All will provide different services across those range of areas. Location is more important for some than others. If you're doing a large language model and it's just training and learning, location is not quite as critical, but certainly for inferencing it is.

We need to be very mindful about the importance overall this plays in underpinning the future direction that businesses will take and where they will grow and their particular needs. When I think of our members, the work that, say, Commonwealth Bank does, the importance—it is now using AI for cyber security, really critical. You need to have very fast times in those sorts of areas. There's a range of different services. These will continue to grow. Location is important, but you can also have it in different areas depending on the technology. We need to be mindful of the value they'll have locally but I would also say there is importance that we can grow this and provide a service to the region too.

The Hon. SCOTT FARLOW: There's been some discussion already about not all data centres are created equal, talking about the hyperscalers, for instance, or those that have sovereign capability here—we need to keep the data onshore. Does the Business Council of Australia take any view with respect to that and whether there should be prioritisation of different data centres here in New South Wales?

WENDY BLACK: They're not all equal. They are all different. I'm certainly not going to say I'm the technical person in all of that. But I think this is why we need to be very open to the advances and the changes that are happening with data centres and the technology that they're using. Therefore, setting strict parameters around prioritising certain ones is going to be very difficult, because it is an evolving system. You've got your cloud compute here, but I'm also doing other digital work here and then I've got my AI and I've got my AI agents. You're probably using all of that in your business going forward, or even in the Parliament here or what I'm doing on my iPhone.

So I think it's very hard to separate them out. I think this is where you have to actually work with the data centres themselves and understand what sort of role is that playing. Am I a hyperscaler? Am I doing cloud? Am I really focusing on AI technology and making sure that we're providing the right environment for the right sort of product that business is going to need? Data centres are not going to set up if they then get restricted in what they can do that doesn't meet the customer demand. I think, as we look forward to what's this new infrastructure technology, we've got to give it some room to allow the real technical people to work out what is critical, what's important, how they evolve and how they meet the demand of the market.

The Hon. SCOTT FARLOW: We heard from the Clean Energy Council prior and I know you were here for part of it, but probably not all of it. Part of their submission is that effectively data centres should be underpinned and matched by the renewable energy rollout, understanding that there may be some timing differences. But if there is a data centre development then that should be underpinned by new renewables that otherwise wouldn't be there. I note in your submission you have cautioned, in a sense, that the rigid coupling of data centre approvals to contemporaneous renewable generation build-out creates investment bottlenecks and risks diverting capital to other jurisdictions. Do you want to outline why you see that being the case and why we shouldn't have a rigid framework when it comes to purchasing power agreements with new renewables that otherwise wouldn't be there?

WENDY BLACK: Certainly. From the outset, we do take a renewables-first approach. As is evidenced by the investments that these data centres have already made—I think it's been quoted several times—over 70 per cent are offsetting with renewables. We agree with that and we agree with the national principles that the Federal Government is setting out as well. However, it goes to the point of how we actually get this aligned. The renewable

energy sector—and I actually endorse many of the comments that I did hear from the Clean Energy Council. We've got some challenges with how that market is operating at the moment.

Some projects are really struggling to get to financial closure. We see this as actually an opportunity and, if you are too strict about saying, "Okay, it's 100 per cent renewables now and you've got to build it now", that is just not feasible. However, if you say, "Look, primarily we want to see renewables first", and there is certainty for these projects to go forward, they will be able to then get some of these projects, particularly some of the wind farms that have really struggled to get financial closure—these are the types of projects that they need to get those contracts across the line.

I see this as a real opportunity to give the renewables sector more momentum from where it is at the minute, because I think some of those projects are struggling. I think, though, that there's got to be more coordination around the planning of that so that timing is aligned, because some of them are not going to be up and running within the time frames when a data centre wants to start. I will take, as an example, Ireland. They have really had challenges with their energy demands from data centres and they have now put in a requirement for 80 per cent, so it's not 100 per cent, which I think is the point that we were trying to make. You've got to be realistic here and be pragmatic about what can and can't be done. But they've got a six-year glide path, which is a recognition of the practicalities of initiating some of these projects.

Going back to our key point, we see this as the opportunity to really kickstart some of the renewables investment that is floundering at the minute in some areas. We can see that there is a win-win situation here. But we've got to be pragmatic about timing around that, what other energy—like exactly with the entire renewable sector. You still need firming under that. The Federal Government acknowledges that there will have to be a role for gas within that but we're saying that's at the lower level. The first objective is renewables, battery storage—the technology around that is increasing all the time—and then a level of firming. You need to have an electricity and energy system that also matches the demand of the sector and what they require for stable energy as well.

The CHAIR: Can I just clarify that? You're publicly opposed to mandating 100 per cent renewables requirement for data centres but at the moment the expected demand in energy consumption associated with data centres will completely blow out our climate targets. We've got legislated emissions reduction targets. Are you advocating that we ignore those targets?

WENDY BLACK: No, not at all. We are very supportive of the targets, particularly the renewables target for 2030 of 82 per cent. We're actually concerned that there are some challenges in hitting that target at the moment because a number of these projects are not getting to financial closure.

The CHAIR: Yes, but there's such huge demand coming in now. If we all look back five years ago when we were looking at this and we were thinking about the trajectory towards 100 per cent renewables, no-one was thinking about this massive extra demand coming in from data centres. If we are to allow that extra demand to come on and to still meet our climate targets, we will need to mandate 100 per cent renewables. How do you balance those two things or make it make sense when they're opposed to each other? You either have 100 per cent renewables and we can meet our climate targets, or we don't.

WENDY BLACK: I think even in achieving 100 per cent renewables, that target actually does—it's a net zero that we're talking about when we go to what is our renewables overall. Even as we get out to 2050 the national figures show that there will be still a level of gas that will have to sit under that for firming. Maybe the technology will completely change and we will get to 100 per cent renewables by then. I don't know, but that's where the modelling is at the moment.

The CHAIR: Where's the net zero coming from? Because we don't have any other technologies at the moment that work to offset that. In the reality of where we are at the moment, what's your solution for how we would get to our net zero targets if we're not getting 100 per cent renewables, at least in the energy sphere?

WENDY BLACK: The net zero is a national target around all the emissions from every part of the—do you just want to talk about the energy sector? Because net zero for the country is the entire emissions of the country not just the energy sector.

The CHAIR: We have climate targets both at a national and at a New South Wales level that because of the demand from data centres—we already have the New South Wales department of climate change having advised the Minister that we're not going to meet our targets. We have the opportunity here to try and turn that around. If we mandated a 100 per cent renewables—genuinely additional renewables—for every data centre, then I think people would be less concerned about the energy demand. But the BCA is saying, "No, don't mandate 100 per cent, but also we still believe in the targets." I'm just trying to work out how you make that work.

WENDY BLACK: The point that I'm trying to make is that I think one of the risks that you have at the moment in not hitting your targets is because there are not people here to actually invest in those projects. That's the biggest problem at the moment. This is why I see the data centres as one of the opportunities where you can get those very long-term, offtake agreements, which will anchor those projects and enable more renewables to come into the system.

The CHAIR: Doesn't that then act in favour of the recommendation that they have 100 per cent renewables then?

WENDY BLACK: But 100 per cent renewables may not always be viable. That's what I'm saying more broadly. Even within the NEM, there is a recognition that you need some level of firming. What we're saying is, yes, priority is absolute renewables and batteries—

The CHAIR: Which are firming. Sorry, when you talk about firming, I'm talking about renewables firming not gas firming.

WENDY BLACK: A battery will not, at this point, provide the level of firming that you are going to need for certainty for a data centre for a period of time, because the technology is not there to last as long as would be required.

The CHAIR: I don't think that's true, but we can get the technical people to tell us that.

The Hon. JACQUI MUNRO: Just on the idea of investment coming to Australia and the benefits that we have as a nation—some unearned, some earned—how does the investment sector business think about the differences between the States? Are you seeing a difference between Victoria and New South Wales, for example, or even Queensland or any of the States? Let's not be restrictive—

The Hon. SCOTT FARLOW: Tasmania.

The Hon. JACQUI MUNRO: I don't know. Is hydropower good for data centres? Anyway—

The Hon. SCOTT FARLOW: Hodgman actually wanted to make them a digital capital. Didn't happen.

WENDY BLACK: There is no doubt there is competition between the States, as there always is between New South Wales and Victoria. Victoria absolutely is trying to win that race, if I put it like that. They have certainly shortened the planning time frames for these projects. At this point the feedback that I get is that Victoria is making it much easier—it is a very attractive place to invest for these types of projects. Having said that, I would commend what the New South Wales Government has done in trying, with the Investment Delivery Authority, to coordinate some of this. I think that is an important step because, as we say in our submission, it is actually pulling all the different pieces together.

There's a great foundation there. We have been critical of the New South Wales planning system but I think this is a step in the right direction, if it can work. It's not just energy; it's the water, it's the telecommunications connections, it is the land use and making sure they're in the right areas. If this authority can help facilitate that—we're not saying remove the standards and the requirements; it's about coordination together, which will actually give you much more longer term certainty. As I said, certainty is key to investment.

The Hon. JACQUI MUNRO: Are you seeing any backlash in Victoria, from a social licence perspective, because of the increased speed of approvals?

WENDY BLACK: Not at this point. Look, there are some minor—there are a few cases for sure. I think because they are largely being built in industrial zones already, in areas which had been manufacturing—those businesses have gone and it's going onto those sites, so they're already deemed to be industrial sites. That is important. We're very mindful that having that social licence and engaging with the community is very important. I was in the US last week, and there is no doubt the data centres backlash is very strong there, and for good reason.

It has cost local communities increased energy costs, concerns about water supply. But what was really evident to us was how much stronger Australia is in that position going forward. We have better foundations. We have a really good connected grid, which they don't have. We have really strong utilities. There, some of the utilities and grid network is very downgraded. They hadn't had any investment in it for a long time—no one wanted to fund it. What you've found is some of the data centres buying up local utilities and actually investing in the grid connections that were much needed, so it's actually turned out to be value.

Likewise here, these companies are seeking to invest and pay their way and bring their energy to the table. I think that will actually facilitate and drive down costs for consumers, if it's coordinated correctly. I know already that Jemena cites a case where it has actually reduced the cost for the local consumers, because the data centre is

taking on more of their share of the costs. I do think there are real positive outcomes for general consumers. I think that will help with the social licence that people can understand that. It's transparent and it's clear.

The Hon. EMILY SUVAAL: I wanted to ask about what you're hearing from industry around some of the additional infrastructure that may or may not be required and, in particular, underwater cables and whether that is something that business is looking at—we've obviously got constraints in terms of how many we've got here on the eastern seaboard—and whether there are any opportunities there.

WENDY BLACK: Yes. It's part of that coordination piece about what is the other infrastructure you need. The cables are very much important to that, also the fibre cables, and that goes to where do you locate these so that you are efficiently using that infrastructure, if you are going to pay to build new infrastructure, and how do you also minimise the disruption to the local community. If you're only having to do one set of cabling through, rather than a whole lot, that is so much better. So, yes, that is an important element that needs to be factored into all of the decision-making, and that's where having a single organisation or authority to look through this and consider, "Okay, how do we put this jigsaw puzzle together so we've got a win-win for everyone?" We come from that perspective.

We see this as a huge opportunity if we have these significant data centres here. Certainly the lesson from the US was that they see Australia as a real opportunity. They said, "Why aren't you doing more of this? You've got so much renewable energy. You've got land. You've got that opportunity. You are a safe place." We had the same message when we were in Japan and Korea earlier in the year as well. They see us as a real opportunity to diversify where they have data centres as well. They just see us as a good place to do business. Our opportunity is to make sure that we do that well and that there are benefits, particularly in, as I said, the renewable energy piece. This could actually be the kickstart, the real driver that we need at this point in time.

The Hon. EMILY SUVAAL: An earlier witness on the Ausgrid panel—I can't remember who it was—suggested that they were supportive of looking at particular zones as a way to help to channel some of this investment and opportunity into the right areas, where there was the infrastructure, or at least potential and capacity. Is that something you would support?

WENDY BLACK: I think it's something that has to be looked at. If there are efficiencies in the infrastructure and you've got underused infrastructure, that would make sense because there'd be efficiencies in that. It makes commercial sense. But you do have to have that commercial lens over it because, whilst you might have an underutilised grid in one part, it may not have water access or it doesn't have the cabling. I think this is where you've got to put it all together and go, "Okay, which area ticks the right boxes? Has it got all of those things? If we're missing one, how could we bring that in?" That's important when there's discussion about how much could maybe go to regional areas as well. It may work in some parts, but it may not in others. I guess that goes back to the earlier question about not being too prescriptive around things, trying to coordinate in a positive way and find solutions. If it's dogmatic and we must do X, Y, and Z, that may not be the smartest way to go about it when there could be more efficient ways to work together and collaborate.

The Hon. EMILY SUVAAL: I want to ask you about the workforce aspect. Your submission talks about the need to grow by around 8,300 people and reaching approximately 17,900 workers by 2030. Have you done any work to try to understand some of the, say, indirect jobs that data centres could generate or some of the other industries that could flow from data centres here domestically?

WENDY BLACK: I think that it's a broader piece. There are the direct, high-tech electrical, cooling and construction jobs. I think Australia is getting an excellent reputation for quality, particularly when we were in the US. AirTrunk is seen as one of the best data centre providers in the world. I liken it to the mining or the resources sector where Australia became the leaders for the quality and the standard that we have and I think we can become global leaders in actually setting the standard, the bar, for this. There's an industry around that. I think also it's more broadly about what does this enable? Think about Silicon Valley. People just tend to gravitate there because there's an ecosystem. Data centres will create that here as well. As I said, it's our way to get a seat at the table in where this technology is going. If you've got companies like Anthropic and then your major ones around Microsoft, Google and Amazon investing here, and their key critical infrastructure is here, Australia gets to have that seat at the table and that builds out the opportunities for other industries and technology industries off that.

The Hon. MARK BUTTIGIEG: Do you have a view on where this is going in terms of technological uptick? We've heard some views that there's a kind of confluence. You've got data centres, which are all about processing power and storage, which requires a lot of cooling, which means a lot of electricity and water. But the technology is improving in terms of chip manufacturing, which means you're getting smaller and smaller processing units requiring less cooling and eventually you'll get to a point where the requirement for all this heavy footprint bulk may attenuate because of that. I just wonder whether or not the Business Council has a view on

whether our real future long term is more in chip manufacturing domestically rather than relying on Taiwan, for example. Has there been any thought on where that's going to end up down the track?

WENDY BLACK: Chip manufacturing is a highly specialised area and that's why Taiwan has really still captured that market. But you do also have significant manufacturing in places like Korea and, of course, the US is trying to also lift its capacity there. That's why Korea is interested in the opportunities for data centres in Australia. I think it would be very difficult for Australia to get into that area, to be honest. This is an ongoing discussion about what are the things that we should have sovereign capability in. I think that would be a very difficult one. How do we make sure we have access to what we need? That is actually having strong relationships with partners and a diversified range of partners as well. But my honest answer is I just don't think we have the capacity and other countries are trying to do that.

The Hon. MARK BUTTIGIEG: Why is that, though, if we've got a highly educated workforce? We're one of the richest countries in the world and, what, we haven't got that ambition? I don't understand. Where does that conclusion come from?

WENDY BLACK: It is a very specialised area and if a company wants to set up and do it here, then there's nothing stopping them doing it and we would support that. But I think there are reasons why it has become centred in particular areas, because of the specialisation of that.

The Hon. SCOTT FARLOW: Taking those points you made about flexibility and not being too rigid, do we need a strategy? Do we need some form of program in New South Wales, or understanding, to outline what our data centre future is going to be, these are the rules that we need around it and this is what we're going to do to attract investment, or is it all just going to fall in our lap?

WENDY BLACK: I think the most important thing is having alignment about the principles and the guidance for how you're going to look at this sector with the national principles as well. That is critical and I think that gives a good guide about how would you then operationalise what those principles are. Because whilst the national Government sets those principles, it's actually the States where the energy gets done. It would be useful to have some guidance but, as I said, some flexibility around it so that you can actually look at things in a holistic way and look at all the different elements that come together to make these projects viable. Because at the end of the day, if they are also not commercially viable, they are not going to come to Australia. I think guidance, some planning—that's also important for the business certainty, because they are looking at long-term investment, long-term energy contracts, and they want that certainty as well.

The Hon. JACQUI MUNRO: That sounds like a strategy to me.

The Hon. SCOTT FARLOW: You referenced Ireland before and what Ireland has done. Ireland and their regime around data centres has been referenced before this Committee. We've also had referenced before the Committee Singapore and their regime around data centres. I can't recall whether Ireland had a moratorium for a period but I know Singapore did while they got their settings right. I'm interested in what your thoughts are on a moratorium and what that would mean if there was a moratorium that was implemented—of course, we're only talking about New South Wales here, so where that would go elsewhere—also, has there been success in other jurisdictions after they've got their settings right, effectively, and your observations in being able to attract additional data centre investment and perhaps build community support for them as well.

WENDY BLACK: Singapore and Ireland are two very different cases from where Australia is at at this point in time. I would not support a moratorium here, because I don't think we're in that same place. We have much better foundations, much better systems already about—things like requiring if you are connecting onto the grid, you have to pay for that already. That's part of the rules here. We have strong planning guidelines, so that's not really what is required here. I think it's a little more of the coordination piece and enabling business to get in and support some of those renewable projects to get away. We're in the catch 22 at the moment—as I said, the ability to get some of those projects away and really give the renewable sector a chance to go forward.

Singapore does not have energy in the way that we do—Ireland, again, not the vast amount of land that we do. They're very different situations. I think Singapore has now been able to move forward, but some of that investment has already gone to other countries, like Malaysia. They did it for their own reasons and they were in a different position. I don't think we're there. If we go through this process in a very smart, clever, thoughtful way—our regulators are also part of this because some of the regulation that we have around connections do not enable for some of this coordination to happen. They haven't been designed for this to consider all the different utilities, so that's where another element is really important as well to make sure that we get this right.

The Hon. SCOTT FARLOW: One thing you reflected on earlier was with respect to the observations that are made to you internationally about how attractive Australia is as a destination. We've heard through this inquiry that New South Wales, in terms of the pipeline, has got the second highest in the world which, I must say,

did astound me. To Mr Buttigieg's point about the smart workforce that we have here, what's the opportunity for us in terms of actually investing more in data centres, attracting more data centres, and the transition of our workforce and our economy into the future?

WENDY BLACK: I think it sends a really clear message to the rest of the world that we support the digital transformation that is going on. I appreciate that some may find that concerning. Those concerns need to be addressed, but this is where we are going. It's a bit like when electricity was developed. Are we going to have some of that or not? We are. This is where we are heading. It's how do we do it in a sensible, measured way. This is where we have a huge opportunity. Those countries are looking at Australia going, "You have got all the things that we wish we had," and you therefore become a leader. Australia becomes that sort of global standard in how we do this and then how do we leverage that, because if you've got that technology here, you do create. You attract all that other investment and you attract the people who want to use those tools to develop the next big thing.

The CHAIR: Thank you so much for making yourself available today. I'm sure there will be supplementary questions and, to the extent there were any questions taken on notice, the Committee secretariat will be in touch. That concludes our time with you.

(The witness withdrew.)

The Hon. ROSE JACKSON, MLC, Minister for Water, Minister for Housing, Minister for Homelessness, Minister for Mental Health, and Minister for Youth, before the Committee

Mr ASHRAF EL-SHERBINI, Executive Director of Operations Resilience, NSW Department of Climate Change, Energy, the Environment and Water, affirmed and examined

The CHAIR: Minister, I remind you you don't need to be sworn, by virtue of your office. Did either of you want to make a short opening statement?

The Hon. ROSE JACKSON: No, Chair, we're okay.

The CHAIR: Minister, when we spoke at budget estimates a little while ago, there was a suggestion that Government would be making a submission to the inquiry. That has now not happened. Could you put on the record why that is the case?

The Hon. ROSE JACKSON: There was a view that the data centre consultation paper that the Government released during the time period in which the Committee was calling for consultations adequately represented the Government's position on where our view on the data centre opportunities and threats and challenges was. Our view was that the consultation paper adequately covered that off and represented our position at this stage.

The CHAIR: Given that it is a consultation paper, do you think it presents a position as such?

The Hon. ROSE JACKSON: No, in a way, but that is because the framework that we might adopt to approach the emergence of data centres in New South Wales is not finalised. In a way, we're not in a position to write, "This is our submission; this is our final position; this is our firm policy at the moment". That's not where we're up to. We're being honest about that. That's why we're doing the consultation paper. In a way, we welcome the opportunity to consider the findings and recommendations of this parliamentary inquiry as well. As I've tried to reflect before, in a way, it would be a shame if we had finalised all of those elements prior to the parliamentary inquiry commencing, or its report, because then the opportunity would be lost to be informed by that work. Yes, two things are happening concurrently here.

The CHAIR: From our position, it would have been useful just to have had some data, even if the submission had said, "Please find attached our consultation paper. We haven't made up our mind on these things, we're consulting, but here's the number of connection requests and the expected growth in demand". It would have been very useful to know some of those issues that only Government has the capacity to tell us. But putting that to one side, I'm very keen to understand, firstly, on the water side of things—because I think in most people's heads at the moment there is the energy and climate impacts, and then there is this very live issue about water connections and how that will impact, particularly in the Sydney Water basin. Are you able to give the Committee any insights into the work you're doing on that? I'm particularly interested in whether there's been any sort of cross-jurisdictional work being done where different water Ministers, maybe in different jurisdictions, are sharing insights. How are we working to try to solve the puzzle?

The Hon. ROSE JACKSON: There has been engagement in the Water Ministerial Council. It was on the agenda at the last Water MinCo meeting. In fact, the Commonwealth's framework or principles paper was tabled then in draft form, and we were able to provide feedback on it, which we did. Whilst I don't necessarily want to go through the entrails of everything that's discussed at a ministerial council meeting, I am happy to say one of the things that I really pushed was I didn't feel as though the original language was strong enough on the principle of existing customers not bearing the cost. That's an example of where we have pushed to ensure that that was a core principle in the national framework.

Subsequent to that, I certainly know that both through the utilities side, so through Sydney Water, and through the policy side, the DCCEEW Water Group, we have been engaging with other jurisdictions. Many of them are in a similar position to us, still in the consultation phase. I think Victoria has come out a little bit further ahead with some firmer framework. Again, we're very engaged with them. I do want to be clear: Our view—the New South Wales Government's view—is that we are a premium location, a premium destination, so we're interested in what the other States are doing, but there is an element of we're not going to be lowering our standards. If other States that maybe don't have some of the competitive advantages that we do, being the biggest, having the kind of capital of Sydney, want to try to make themselves more attractive by lowering their standards, that's not necessarily an approach that we want to take. We are engaged with them, though.

Just more broadly, Ms Boyd, I'd say in terms of the things that we have been doing, from the water side, we have been pushing those key principles to inform part of our approach going forward. Existing customers don't bear the cost. This is going to be expensive at times and in parts, and it is not fair that existing water customers

pay for private industry growth. That is a core principle: preference for recycled water. It's not new information that the potential demand is significant. We are anxious about that. We don't want to put Sydney or our regional towns under pressure for water security—so a preference for recycled water, again, a core principle that we would have, and clear data and information and reporting from data centre proponents and users. Often it can be challenging for us to get the kind of information that we want. We want to make sure that we're really clear up-front about our expectations on that.

One of the last ones, although there are other things, is what to do about extreme events, mainly drought, so drought preparedness. We want to make sure that we're not just planning for the average years; we're planning for the extreme years. In terms of your question about what have been the general principles that have guided our engagement so far, that has been it. You can see some of those principles reflected in the consultation paper, and we've been working behind the scenes—and when I say "we", that's DCCEEW Water Group and Sydney Water mainly—on fleshing out what that might look like as the regulatory approach is finalised.

The CHAIR: Thank you, that is incredibly useful information. The Water Services Association of Australia that we had in last week I think was pretty critical of the consultation paper. I'm sure they'll be feeding in separately anyway, but they stated that the consultation paper mentioned sustainable water use but doesn't yet propose any actual measures. I understand that there's currently no data centre using recycled water. There's significant concern that we're already approving data centres without the appropriate protections in place. Is there anything that you think can be done to go back and fix those things after the event? If we do end up with a set of really good principles around the sustainable use of water, is there a way to back-solve for that with the ones that have already been—

The Hon. ROSE JACKSON: I'm not sure in terms of whether their planning approvals or other approvals provide an opportunity for us to revisit those questions. I'd have to check that. I would say that I am aware of some of the things that WSAA has raised. I had a specific round table or dialogue or meeting with the New South Wales branch of the Australian Water Association about some of these issues as well. I'm very engaged with wanting to make sure that that feedback and that dialogue is there, so I am familiar with those concerns. I do want to reassure them and the Committee that we are coming in at the pretty early stages here. I accept that there are some data centres that have been approved and are operating but, really, we are at the very early stages of the growth of the potential impact of this emerging industry. Ash, you will correct me if I'm wrong but, on currently approved and operating data centres, we're at less than 1 per cent of water demand.

ASHRAF EL-SHERBINI: Correct.

The Hon. ROSE JACKSON: It's pretty small. It's pretty limited—not to say that we don't recognise that the growth potential is there. But to your question about have we already missed the boat or is there this large group of data centres that are out there operating, there are a small number, but the growth is really into the future. That's why this work is happening now.

The CHAIR: That's really useful. I'm just going to ask one more question on water, picking up on something you said about drought. You talk about how we're at the beginning, and I think part of the issue that people in the community have had is that data centre development just suddenly appeared for a lot of people. You're right: For years we've had different little data centres around the place—they have been there—but now we're seeing these at hyperscale and more people are experiencing a proposal on their street or whatever it happens to be. I think it's collectively coming into our consciousness, and part of the reason for having this inquiry was for us to better educate ourselves, the public and the media about exactly what was going on.

One of the issues that I've been grappling with in this inquiry is working out what is critical and what isn't, and how that then plays into how we manage resources. If we have a bunch of data centres and we don't know which ones are necessarily critical in terms of housing government data, health data, banking data or whatever and are just hosting cat videos—ideally, if you knew which ones were which, you could say, in a drought, you can turn your cat videos off, but we need these ones. Do you think there needs to be built into the framework some sort of criticality element to enable, from a water resources and even energy resources perspective, the ability for government to react in a better way?

The Hon. ROSE JACKSON: Absolutely. I think you have well articulated one of the challenges there, and that goes to the point that I made about data centres sharing data or sharing information, because I completely agree with you. To the extent that this is a real opportunity for our State to lean into future economic development and productivity, I'm for that. I don't think you can have your head in the sand in terms of the immense opportunity that these kinds of technologies present our economy. I am for that, and to the extent that that is underpinning, as you say, critical civil aviation and health—those things that keep our planes in the sky or keep our hospitals running—they are things that we cannot turn off without massive economic impacts. But to the extent that it's

ChatGTP or Character.AI, clearly it's less critical. So, yes, as part of our drought planning framework, it is about having a system where you are able to determine what is indeed critical and what is not.

Obviously, proponents are somewhat protective of that information at times. They would argue, perhaps rightfully, that it's commercially sensitive and those sorts of things. But my view would be not too dissimilar to yours, which is, to the extent that we have drought planning frameworks in place, part of that is saying, at this point, we do need a more robust engagement on what it is that this piece of infrastructure is sitting behind, and is that critical or not? Difficult decisions have to be made in droughts at an individual level, by government and by industry. This is going to have to be one of them, and we just have to get in early to have a clear and agreed framework for how to make those decisions. I would just say briefly as well that I think it is important to emphasise that your point about the lumpy nature of data centre growth is a very valid one too.

This is a struggle for us in terms of the way that we do the financing. Most of the financing models that we have through Sydney Water assume gradual or somewhat linear growth—household growth, industry growth, the growth of our city—whereas this is not that. It is suddenly emerging as a very large demand centre. The lumpy nature of that is distinct to data centres and does present challenges in terms of how our traditional models of growth are not well suited to it. That's why this piece of work is happening. We're trying to lean into accepting that our business-as-usual models don't really work in this circumstance.

The Hon. SCOTT FARLOW: Thank you, Minister, for your attendance here today. Before the inquiry we've heard evidence from several witnesses—at least they have repeated the statistic—that if all data centres were to be developed, that 25 per cent of Sydney's drinking water would be used. Is that similar to what the Government is modelling as well? Or is it nowhere near there?

The Hon. ROSE JACKSON: That is obviously Sydney Water modelling. I'm familiar with it. I also know that there are some people who think that that is not accurate or have concerns about it. All I would say is I meet with Sydney Water regularly and I know that they're engaged with people who want to test that modelling, want to challenge that modelling and have different views on it. I do rely on them to provide me information about what pipeline they're seeing. I have to. They're the water utility. They're the ones that have to plan for the connections and the yield data. I have confidence in them. I think that the information that they provide is sound and is well developed. To the extent that people have questioned some of those figures, Sydney Water is engaged with, is talking to those people and is working through a shared understanding of what that would look like.

The Hon. SCOTT FARLOW: Some of the submissions that we've received in terms of addressing that—one was from Data Centres Australia, who suggested the use of purified recycled water but complained that effectively that wasn't being made available by Sydney Water. Does the Government have any proposals to start using purified recycled water, particularly for data centres or industrial clients?

The Hon. ROSE JACKSON: Yes. I'd say this is an example of where there are opportunities in data centre growth, because actually it's true to say that the take-up of recycled water for industrial and environmental purposes has been sclerotic. That's been a missed opportunity for our State. There are a number of industrial and environmental uses for which recycled water could be used, taking pressure off the drinking water—potable water—demand. But unfortunately, in a way, that hasn't been something that has really been a focus. So it's a good opportunity, in a way, with the data centre growth, to really lean into that. The problem is that costs money. That recycled water infrastructure is not cheap, both on the plant side—the treatment side—and the connection side. That's all money.

This goes back to the point about who pays for that. Who funds that? If your question is, "Is the Government going to fund the development and construction and rollout of a massive recycled water program for industrial customers like data centres?", no, we're not. We're not going to pay for that because that's Sydney Water customers cross-subsidising this industrial use. We could not be clearer that that is inconsistent with the framework that we have adopted. The good news is that the proponents themselves are actually interested in and acknowledge that this is a cost that they will have to include as part of their planning. So I do think we have an opportunity to invest in and deliver more recycled water for industrial purposes here, but we're going to need a framework in which the developers—the proponents—fund that. That is an open conversation, and they recognise that that's part of what they're going to have to do if they want to be part of our market.

The Hon. SCOTT FARLOW: Minister, to that point in terms of that framework and that discussion, what are your milestones in terms of being able to develop either some sort of private market sounding or some form of memorandum of understanding with Sydney Water and those developers to be able to bring online new purified recycled water facilities?

The Hon. ROSE JACKSON: There are two key things that need to happen. I would say the time frames over the next six months align with the Committee's reporting timetable. I don't have firm dates available. The

Cabinet needs to consider the feedback on the consultation paper and translate that into some kind of policy framework. The purpose of the consultation paper is to inform the finalisation of a policy framework, which speaks in more specific detail to the questions that you're raising. The consultation paper and high-level principles need to become a firmer policy road map.

That is a key milestone that is to happen over the next few months. The other critical piece of work alongside that is Sydney Water needs to grapple with and finalise this question of financing models that I've mentioned is unresolved because the traditional developer contributions program that Sydney Water has to support housing and other infrastructure growth of that nature is not well suited to this particular challenge. Sydney Water recognised that and are engaged in what a different model to support industry contribution to water infrastructure would look like for data centres.

The Hon. SCOTT FARLOW: The other suggestion we've had before the Committee has come from the Sydney Desalination Plant. They have said that they could double their capacity to be able to increase supply for data centres and generally for additional supply in Sydney. Is that something that you're entertaining?

The Hon. ROSE JACKSON: Absolutely. Everything's on the table. I've met with the Sydney Desalination Plant as well; I visited it. I'm well aware of the fact that you could double its capacity. The concrete pads are right there. It's an obvious opportunity. Again, that costs money. It's billions of dollars to double the size of the desal. It's currently operating permanently.

The Hon. SCOTT FARLOW: And energy as well.

The Hon. ROSE JACKSON: That's right. The current operation of the Sydney desal, as I'm sure you're aware, is energy offset. When you drive down to Canberra, the wind turbines that you see there are offsetting the energy of the Sydney desalination plant. But it'd be a \$1 billion-plus project.

ASHRAF EL-SHERBINI: At least \$1 billion.

The Hon. ROSE JACKSON: It's with a B isn't it, Ash?

ASHRAF EL-SHERBINI: Yes.

The Hon. ROSE JACKSON: Someone has to pay for that. Now that could lift the rainfall independent water yield in Sydney from 15 per cent to 30 per cent—not insignificant. I do entertain the idea that that is something that Sydney might consider into the future. Of course, the planning was done in the original proposal but, when things cost that kind of money, it's an obvious question of who's going to pay for it and who gets the benefit of that, so that would be an example of what we need to work through. Are existing Sydney Water customers going to fund that? Is it really being done for their benefit? If so, that's one pathway. But if it's being done for the benefit of private industry—not that that's necessarily a bad thing; we do want private industry here in New South Wales—but that does change the consideration of the funding models. That's the type of thing that I was referring to when we're talking about how we consider the financing of these kinds of big water infrastructure projects.

The Hon. JACQUI MUNRO: On that, Minister, are you actually considering the financing models on those projects?

The Hon. ROSE JACKSON: Sydney Water is working through what alternative or different or updated financing models it might adopt to ensure that data centres are able to access water, but that it is not paid for by existing customers.

The Hon. JACQUI MUNRO: What's happening at the moment with approvals for data centres?

The Hon. ROSE JACKSON: They're going through Planning. Obviously as part of that, like any industrial proposal, they have to provide information about their water demand, energy, those sorts of things.

The Hon. JACQUI MUNRO: Are any agreements being made at the moment that sit outside of the traditional agreement between Sydney Water and other private connections?

The Hon. ROSE JACKSON: I'm not entirely sure. I'll have to ask Sydney Water. I can take it on notice, but I think you had—

The Hon. JACQUI MUNRO: Mr El-Sherbini was nodding.

The Hon. ROSE JACKSON: Do you know if Sydney Water have finalised any of those? I'd have to ask them for the detail. I don't know it.

ASHRAF EL-SHERBINI: I can provide some comments on that. Generally, for the ones that have gone through, as the Minister was referring to, there's an infrastructure contribution charge that applies in a local area

where it's being developed. On top of that, there's an additional charge of any connection to the pipeline that that data centre needs. As per the Minister's comments, what Sydney Water is working on at the moment is a model that reflects the fact that data centres require a different scale of water demand. They require a different level of reliability and a different quality. In order to meet that, the pricing model has to change.

So, yes, Sydney Water has taken some steps towards identifying what that looks like. It's through a process that is called a developer planning agreement process. Sydney Water receives a high number of inquiries—up to 90 inquiries just in the last 12 months—for data centres. The next step is to work through that in some level of detail to actually identify how valid they are, what their specific needs are and how much it's going to cost to satisfy that need. That goes to answering your question. There have been seven of those agreements entered into between Sydney Water and data centre proponents. That agreement allows Sydney Water to get more granular information about that data centre's water demand and their needs, and then develop what the infrastructure solution is for that data centre and, therefore, the cost.

The Hon. JACQUI MUNRO: Is that essentially a first in, best dressed situation? Without an overarching strategy or plan, how does Sydney Water determine future need and the prioritisation of approvals?

The Hon. ROSE JACKSON: You'll have to ask Sydney Water that.

The Hon. JACQUI MUNRO: Have you asked Sydney Water that, Minister?

The Hon. ROSE JACKSON: We're very engaged with Sydney Water. Sydney Water witnesses were here. You can put questions on supplementary notice.

The Hon. JACQUI MUNRO: But you're the Minister responsible for water and Sydney Water report to you. Do you have that information?

The Hon. ROSE JACKSON: I don't have details and information about how they are triaging the 90 applications that they have. I don't have that on me. If you want detailed information about how they are triaging and managing applications, you can ask them.

The Hon. JACQUI MUNRO: Does it concern you that there is no data centre strategy that will be provided?

The Hon. ROSE JACKSON: Obviously it does, which is why we're developing one.

The Hon. JACQUI MUNRO: So you are developing a strategy, is that correct? Because we heard evidence from Simon Draper last week that there would be no data centre strategy.

The Hon. ROSE JACKSON: The entire purpose of the consultation paper is to finalise a policy framework for data centres. I don't know whether you want to call that a strategy, a policy framework, a road map.

The Hon. JACQUI MUNRO: He was pretty clear. He said there will be no data centre strategy and that there would be general guidelines proposed for data centre proponents.

The Hon. ROSE JACKSON: I would argue that guidelines for data centres represent a data centre strategy. Honestly, I feel like this is a semantic debate. The reason that we're doing the consultation paper, the reason—

The Hon. JACQUI MUNRO: But, Minister, there's a difference between guidelines for data centres to comply with or generally agree with or have a prioritisation level according to those guidelines, and a strategy that looks at different statewide precincts, for example, or suggests areas that might be more effective for connection to electricity and water. They're pretty different concepts.

The Hon. ROSE JACKSON: I don't really accept that they are.

The Hon. JACQUI MUNRO: On the 25 per cent of Sydney Water's water being used for data centres, is that the figure that you are using when you're thinking about planning for guidelines to give to data centre proponents?

The Hon. ROSE JACKSON: All of the different modelling is being considered as part of that, yes, absolutely.

The Hon. JACQUI MUNRO: So you're not relying on Sydney Water's modelling?

The Hon. ROSE JACKSON: Yes, we are relying on Sydney Water's modelling. But we also recognise that other people have raised questions about it. Their questions and feedback are being considered as well. That's what the consultation paper is intended to draw out.

The Hon. JACQUI MUNRO: So IPART has dismissed 25 per cent. I'm just trying to understand how you're making decisions about what happens with data centre operators as they're coming through the pipeline at the moment. Is the Government going to come up with its own number, separate to Sydney Water? Is that what you're saying?

The Hon. ROSE JACKSON: I don't anticipate that we will. We will consider Sydney Water's feedback. But, as I indicated, they are engaged, as the water utility, with considering the information that has been provided about whether that is a reliable figure or not. That's something that they are working on.

The Hon. JACQUI MUNRO: So you're anticipating that the figure will change, I presume?

The Hon. ROSE JACKSON: I'm not anticipating anything. I am open to the feedback that is received on the consultation paper.

The Hon. JACQUI MUNRO: When was the first time that you were asked about data centres as a consideration of future policymaking? When did data centres first come across your desk as something that you needed to be thinking about in more than a one-off approval process?

The Hon. ROSE JACKSON: I am obviously unable to recall that off the top of my head.

The Hon. JACQUI MUNRO: Are we talking last year or two or three years ago?

The Hon. ROSE JACKSON: Some time ago—some years ago. I honestly don't recall. I just have to indicate that I'm here till 12.00 p.m. I have a ministerial council meeting for building, planning and housing at 12.00 p.m. So I will take on notice when the issue was first raised with me. I cannot recall when it is and I apologise that I have to go.

The CHAIR: I apologise, Minister, for not realising that we had got to 12.00 p.m.

The Hon. ROSE JACKSON: That's okay, Chair.

The CHAIR: Thank you so much. We really do appreciate you being here. Those questions that were taken on notice and also supplementary questions that will be forthcoming will be spoken to you about by the Committee secretariat. That concludes our time with you.

(The witnesses withdrew.)

Dr SUE KEAY, Director, University of New South Wales AI Institute, affirmed and examined

The CHAIR: I now welcome our next witness. Would you like to commence with a short opening statement?

SUE KEAY: Thank you for the opportunity to appear today. Let me start with how data centre investment intersects with our ability to conduct scientific research. Every researcher is now a GPU user. AI is becoming foundational across all scientific disciplines. Researchers who cannot access compute in the form of GPUs cannot compete internationally. UNSW and the University of Sydney are both global top 20 universities, and keeping this rank requires attracting and retaining world-class researchers. That means giving them access to world-class research infrastructure.

Without secure, affordable, long-term access to GPUs, New South Wales will lose talent, lose research investment and lose commercialisation opportunities to jurisdictions that do provide access to compute. That compute and its scientific and economic benefits are controlled by the tenants of the data centres, not usually the data centre operators, who are the ones who are obtaining planning approvals. The tenants, hyperscalers, neocloud providers and enterprise customers are the ones who control the compute. They decide the use case, the workload, the data governance arrangements, the type of backup power systems and cooling suitable for the architecture of their systems.

The conditions New South Wales places on data centre operators have limited impact on the tenants. In Ireland tenants of data centres are required to enter demand flexibility agreements as a condition of grid connection. Some US county-level agreements require operators to flow conditions down to tenants through their lease agreements. In the absence of public investment into the GPU infrastructure, we need to build AI capability and to support scientific research. We could require data centre tenants to give access to GPUs on favourable terms to ensure that the value and benefits remain onshore and balance the profits that will be taken offshore.

The value of this contribution to uplifting New South Wales' AI capability and economy can be measured by requiring reporting on things such as GPU hours delivered to public interest users, Australian AI jobs created, datasets activated safely, and local supply chain spend. New South Wales has a genuine opportunity to build on its leading digital infrastructure. Data centre investment is coming to Australia, and the opportunity is to shape that investment to deliver lasting public value. I invite the Committee to visit the UNSW AI Institute if you'd like to see some of the research that we're conducting that is reliant on access to large clusters of GPUs and the value that this unlocks.

The CHAIR: Thank you very much. I am going to ask one question and then I will throw to my colleagues. It's a little bit nebulous, but bear with me. One of the issues that I see in this space is an abundance mentality. I think we've had this discussion with other witnesses at different points in this inquiry. All the time, proponents of data centres see this unlimited water, unlimited energy, "We can just grow, grow, grow". We are doing ourselves a disservice in terms of incentivising efficiencies. How do we do that in a way where we are still attracting good-quality data centre proponents but we're not hindering the advancement of technologies behind the scenes that would hugely improve and make their processes far more efficient?

SUE KEAY: I'll talk about that from two aspects. From a research organisation perspective, there's not access to the abundant compute that's being built. I think there is sometimes a misconception that just because there is a high level of data centre investment and there is AI infrastructure being built, somehow that's accessible to Australians, because it's not—not unless we pay for it. The other thing I would say is that this is quite an opportunity for Australia. If there is anything that we are good at in the artificial intelligence space, it is building artificial intelligence for resource-constrained environments. This is exemplified by us leading the world in remote operations in mining, for example; us being the first country in the world to automate our ports—all of these things.

We've had to specialise in technologies where they often have to operate in very remote conditions where they have no access to power, where they have no access, often, to communications, and where they often have to operate in extreme environments of heat, so they're quite robust and reliable. I think that's actually an area where we could make a significant contribution. But at the moment, to develop a lot of the AI models that would unlock value from many of our public datasets requires access to GPU clusters. There will be these GPU clusters built in Australia, but with no public investment to either buy the infrastructure itself or buy access to the infrastructure.

The CHAIR: The other issue that has been perplexing me is how much sovereign capability we require and what we mean by "sovereign capability". Instinctively, when I think about having sovereign capability, it's taken to the extreme—if we were in a war scenario and we needed to continue to operate all of the critical things

our society relies on, that we would be able to continue to do so because we would have control over where that was housed. But then I also know that a lot of this data doesn't live in any particular place—it lives all over the place—and that we don't necessarily know what each of these data centres is housing and how much of that information is flowing into different countries anyway. Other people talk about sovereign capability in terms of what we can then export. What is sovereign capability, from your perspective, and how much of that actually matters?

SUE KEAY: The term "sovereign" means different things to different people. From the Federal Government perspective, it seems to mean anything that we can get our hands on in Australia. From my perspective, it is something that we own and control and that we are getting the profits from. In terms of where we can compete, I don't think anyone is suggesting that we should be manufacturing our own chips. Fortunately some other countries, like the UK, are looking to do that, so hopefully we will be able to access more vendors to be able to buy chips in the future. However, in all of the other layers of the AI stack, I think we can and should compete. Australia to date has been relatively neutral in being able to retain our AI talent, as I said, but if we don't give them access to the infrastructure, that is likely to change.

While we have that AI talent, we have the capability to be building our own AI models. I'm not suggesting that we build a frontier model that would be competing with Anthropic or OpenAI, but the direction that AI is heading is that we are now starting to see more value out of domain-specific AI models. Many of those are well beyond large language models—they are multimodal models. They would be able to model from sensor data and include all sorts of different data, like temperature and moisture, that we would be able to interrogate to develop solutions. Another good example of that would be in the healthcare space where researchers are actively using data to be able to develop models so that we can design precision medicine and look for new drugs—use it for drug discovery. There are a whole range of opportunities that open up if we turn our minds to how we can be developing our own AI models.

Unfortunately, compared to peer nations, what we're not seeing is public investment in the infrastructure to build those models as well as public investment to free up researcher time to develop those models. Other countries, like Norway, have schemes where they actually fund the development of AI models for research purposes. At the moment, this is something that researchers in Australia have to cobble together either in their own time or as part of other research grants. There is no overall plan for our valuable data sets that we need to be processing onshore and keeping closed from other countries or how we would go about building our own AI models.

To your point about sovereignty, people may have mentioned the US CLOUD Act. Just because we have data housed onshore, if it is with a US tech company then that doesn't necessarily mean that other countries might not access that. There is a solution to that, and that is that we have alternatives to the hyperscalers. We have Australian-owned neocloud providers. We could be making a deliberate choice about where we are building our models. Certainly in the application layer we already have a lot of activity. UNSW is the university that produces the most startups, and increasingly they are all AI startups in some form.

The Hon. MARK BUTTIGIEG: Could I just follow up on that. Conceptually, your point that we need stimulating research in developing AI models, why is that necessarily a function of the degree to which we develop data centres, for example? In an interconnected world, theoretically, you can grab data inputs from anywhere, right? Because everything, just like the international banking system, is interconnected. I want to get a feel for why it's so critical to have stuff domiciled here.

SUE KEAY: We can't exchange our health data anywhere. There are regulations around how that data is stored and treated. It's actually not possible for it to just be part of an interconnected network. It would always have to reside in Australia and undergo quite rigorous scrutiny before it is opened up for use in a model. Going back to your question about interconnectedness, the challenge at the moment for NSW Health, for example, is where would they get access to GPU clusters to be able to build their own AI models?

The Hon. MARK BUTTIGIEG: Does that then presuppose that, eventually, we're going to have to end up with these islands of data centres and processing capacity within nation-states?

SUE KEAY: No. I think you need to have some sort of interoperability and ability to be able to make things portable between locations. I'll give you an example of how this works at UNSW. We have researchers who have their own GPUs. We have our own supercomputing facility, Katana, which is located onsite. We also contribute to one of the nation's two supercomputing facilities, the NCI, National Computing Infrastructure, in Canberra, and we run a lot of workloads. We have three times more the demand for those facilities than we can meet. Our demand signals are suppressed because, in some cases, our researchers will go overseas or work with international collaborators to get access to the GPU clusters that they need to get access to. But, fundamentally, they're not able to do that in an area like health because they can't transport that data. In collaboration with the

University of Sydney, we're also looking to invest in what we call GPU as a service, where we work with a local Australian neocloud provider, who will maintain and have the most up-to-date GPU infrastructure, and we pay to access that infrastructure, rather than paying to own the infrastructure ourselves. So I think it's going to be a hybrid model.

The Hon. MARK BUTTIGIEG: Depending on whether or not the data is required to be sovereign, like in the health space.

SUE KEAY: No. That particular model—because it's an Australian neocloud provider and it's all onsite in Australia—would meet the requirements for even sensitive workloads.

The Hon. MARK BUTTIGIEG: Do you have any sense of what's happening with the technology—this has been a theme of the evidence today—in terms of microprocessing, and the ability for throughput and microprocessing to get smaller and smaller and require less and less cooling and energy? Where is that going in terms of chip manufacturing? What opportunities might there be for Australian manufacturing in that area? Rather than just building AI models and doing all the front-end stuff, the back-end supply—does the university have any insights into where that's going?

SUE KEAY: Probably the New South Wales semiconductor bureau would be best placed to answer that. But my understanding is that we are looking at building some of our own chips with niche capabilities here in Australia. But unless we're prepared to invest in a fab—which would cost, I think, to the tune of \$100 billion—it's unlikely.

The Hon. MARK BUTTIGIEG: A hundred?

SUE KEAY: I think it's about \$100 billion, but I'd probably have to take that question on notice to get the full figures. We're not going to be creating semiconductors at scale that would be adequate to be able to build our own GPU, if that makes sense.

The Hon. EMILY SUVAAL: Can I just ask a question, Chair, just to follow on from Mark's? Dr Keay, earlier in the inquiry—I can't remember his name—an academic from University of Sydney that was here mentioned some of the work that UNSW was doing around digital twinning. It's not mentioned much in your submission—which is very excellent, by the way—but what are the opportunities for that sort of work and partnering with government in terms of trying to realise some of the opportunities? I live in the Hunter Valley, so it's a keen interest for me to understand how our part of the world can be a player in this.

SUE KEAY: There are enormous opportunities. It's a bit hard to know where to start. You can have digital twins of just about anything, including the human body. Potentially in the future, we will have our own digital twins that we can use to help discover medical treatments that are best suited for us as individuals. So, yes, there are a number of different ways that that could be applied. In an area like the Hunter Valley, I think one of the challenges is around how to ensure that a lot of our industries are making sure that they are actually getting the value out of the data that they already own and are able to leverage that to their own benefit, rather than to the benefit of the companies—for example, on a farm, the agronomy company that comes in and will do a drone survey, collect all the data and then take that and make additional value from that data without any of the benefits going back to the landholder, who you would argue the data actually belongs to.

It depends on the company. There are a lot of different arrangements now. There is a lot more value in our data. This is not something that maybe the inquiry is getting into, but a lot of the challenges against Australian copyright regulations, for example, I think, indicate that other people recognise the value of the data that we have in Australia. At the moment, we're not really recognising the value of our data—our data as individuals but also for many of our companies, the data that they hold as well, if that makes sense.

The Hon. EMILY SUVAAL: Yes, it does. We had the Writers' Guild in an earlier session, and that certainly came up as a theme.

The Hon. JACQUI MUNRO: Thank you so much for appearing today and for your really helpful submissions. In your opening statement, you spoke about tenants. That's something that we haven't really explored during this inquiry so far, but I think it's a really interesting part of this idea of a data supply chain, in a way. You mentioned Ireland. Are those requirements for tenants of data centres standard across the country, for Ireland?

SUE KEAY: I'd have to take the question on notice. I don't know.

The Hon. JACQUI MUNRO: Are you familiar with how data centre operators and tenants work together in that environment? Is it that tenants directly deal with their requirements under law, through government, rather than interacting with data centre operators?

SUE KEAY: I think the negotiation is directly between the electricity companies or the grid operators and the tenants, as opposed to the data centre operators, because, in many cases, the data centre operators have no control over what workloads their tenants are putting onto that computing infrastructure. It's very hard for them to be able to guarantee that there's going to be continuous load because it really depends on what the tenant is putting to that infrastructure, what sort of work.

The Hon. JACQUI MUNRO: My other question was around international agreements like Horizon Europe and what kind of opportunity they offer us. Before, I think you suggested that we're basically missing out on research discoveries because we don't have enough compute power in Australia, or at least sovereign compute power. What opportunities do agreements like Horizon Europe offer to us?

SUE KEAY: It's going to open up a lot of opportunities if we're able to access some of that funding. We already do have a relationship, I believe, that gives us some ability to access Euro HP, but it's on a competitive basis.

The Hon. JACQUI MUNRO: In terms of using tenancies as a way to encourage GPU use for research and science and technology, how would you suggest that we put that into a framework?

SUE KEAY: I'm not sure whether this is a State or Federal government thing, but when considering the pros and cons of foreign direct investment, at that point I think that's the conversation that needs to be had about how there can be—I mean, this is in line with the Federal Government's expectations of data centres and AI infrastructure suppliers.

The Hon. JACQUI MUNRO: Through SOCI or—

SUE KEAY: I don't think they proposed a mechanism in that recommendation, but they did suggest that there should be—I can't remember the terminology—something along the lines of discounted rates for researchers, startups and SMEs to get access to computing capacity, but it was silent on the mechanism by which you would do that. I did have a quick search to see what other jurisdictions do, and I think the challenge for Australia is that in other jurisdictions it's public-private partnership, and there is a level of public investment. If you're trying to do this with zero public investment, then you do have to find mechanisms that would require or incentivise the private providers to give discount access.

Now, that could be through anchor tenancies, offtake agreements—there's probably a variety of mechanisms that could be used. But I think the key distinction is that it has to be nuanced, that it's not the data centre operator in most cases. There are some—I don't know whether this is the right term—vertically integrated data centre operators who also own the compute that's in their data centre, but that's increasingly unusual. The data centre operators tend to just be providing facilities for their tenants, and it's the tenants who control the compute and, importantly, are the ones getting most of the economic benefit of this whole AI boom.

The Hon. JACQUI MUNRO: So, basically, it's unusual to find governments or equivalent comparable governments at this point in time not making public investment into data centre infrastructure.

SUE KEAY: That's correct, in such a way that it would free up that capacity for research and startups.

The Hon. SCOTT FARLOW: If I could follow on from that question, we are in this prism where data centres are somehow new and that this is something that's caught us unaware: We didn't see the rise of data centres. Is that a fair characterisation in the Australian market or is this fairly mature?

SUE KEAY: Yes, I guess there are data centres and data centres. The new version of data centres are going to have GPU clusters, which would not have been at all common in the past, so it's quite a different architecture and quite different requirements, particularly from an energy perspective, because AI inference and AI training are what is absorbing a lot of the energy and then requiring the cooling. That level of compute demand was not there previously.

The Hon. SCOTT FARLOW: But then if you go to the Singaporean example or the Irish example, in terms of their arrangements this was causing, at least, policymakers in those countries to turn their minds to this issue some time ago. Was that because there was an exponential demand or growth that occurred in those countries that wasn't seen here at that stage, or have we been caught napping a little bit?

SUE KEAY: I don't know the answer to that, but the observation I would make is that many of our peer nations, like Canada, have had a national AI strategy since 2017 and so have been thinking about this for at least eight years. We still don't have an AI strategy in Australia. We have an AI plan, but when you articulate—

The Hon. SCOTT FARLOW: To that point, is there a distinction between the plan and the strategy? I think you were here for the evidence of the Minister, who said, effectively, it doesn't matter if it's a strategy, a framework or a plan.

The Hon. JACQUI MUNRO: Or if it's guidelines.

The Hon. SCOTT FARLOW: Or if it's guidelines. Do you think there is a difference?

SUE KEAY: Yes, and I can tell you why. That's because in a strategy, you are articulating what your ambition is and where you want to be in the future, and then you're setting up a plan of how you would get there. A good example of this would be South Korea; I think their strategy was put together in either 2017 or 2018. They put forward the proposition that they wanted to be the number three AI country in the world. That's a very bold ambition, because number one and two are US and China. South Korea has a population of 51 million, but it has a smaller GDP than Australia. To that end, in their strategy they actually went department by department in their government and articulated what were the actions that would need to happen in each of the government departments to help to achieve that strategy, and they've backed it up with investment.

Recently, South Korea announced the equivalent of A\$112 billion worth of investment into AI. Their government has secured 25,000 GPUs through NVIDIA, and that is just to encourage industry to purchase another 225,000 GPUs for South Korea. Because they see it as critical to their economic survival and for industry, they are investing accordingly. They've also invested in a program to encourage 10,000 AI startups. So other countries are treating this quite differently and have been thinking about it for a long time. I have no real insight into why that has not happened in Australia.

The Hon. SCOTT FARLOW: Just on some of the points around planning that you raised in your submission and the clustering concern that exists, you noted that you'd need precinct-scale master planning to be in place rather than looking at just individual site-specific plans. Can you expand on why that's important and what the benefits will be? Potentially, if we weren't to take that approach, what could the detriments be?

SUE KEAY: I'm probably not the right person to answer that question, but certainly we could get one of the other authors to come back to you with that.

The Hon. JACQUI MUNRO: I know the submission by the UNSW Institute for Industrial Decarbonisation was hyperlinked against your name, but I note the distinction between the UNSW AI Institute and the Institute for Industrial Decarbonisation for the record.

The Hon. SCOTT FARLOW: My apologies in terms of my references to the UNSW AI Institute submission.

The Hon. EMILY SUVAAL: Mine too.

The Hon. SCOTT FARLOW: Thank you for that point of clarification, Jacqui. It's most helpful and will probably take away all the other questions that I might have had. I will ask you this one that I did have from that submission, but I think you have provided some insight into it as well in your answer to Mr Buttigieg before. Back to the point of \$100 billion for a fab, as we've seen, Taiwan is the leader when it comes to semiconductor development. Jacqui and I have both visited a foundry, and even between foundries in Taiwan there is a huge difference. What do you think are some of the areas where we potentially could have an Australian domestic manufacturing capability that could go into the data centre process? In that UNSW AI Institute submission, there was some talk about things like cooling, power and modular infrastructure. Are there some other areas that might not be at the forefront of what we think of in terms of data centre technology, but would be fundamental to support data centres that we could produce in Australia competitively, and we could have our own industry to support?

SUE KEAY: It's not something I've thought in detail about, apart from—I should also mention that I'm the chair of Robotics Australia Group, which represents the robotics industry. We have been advocating that it would be beneficial if we could encourage the people who are constructing data centres to look at how they are incorporating more automation and robotics in construction, because we're going to be coming to a building crunch where the Brisbane Olympics, all of this data centre investment and 1.2 million homes are all going to be competing for the same pool of people.

We have a lot of these technologies developed in Australia but are struggling to get the customers to be able to grow and scale. Any ways that the uptake of technology could be incorporated in the construction of data centres would certainly be very helpful. In terms of what technologies Australia could be producing that are incorporated in the computing infrastructure within the data centre, I have less of a view, apart from I know that there are some investigations into how, for example, you would automate a lot of the cable-laying trays within data centres. But, again, that's probably more on the construction side rather than the ongoing maintenance and operation side.

The CHAIR: We are out of time. I think you'd make a fabulous dinner guest.

The Hon. SCOTT FARLOW: Is that an invitation?

The CHAIR: I'd just be picking your brain the entire time. That does conclude our time with you. To the extent that there were questions taken on notice, the Committee secretariat will be in touch.

(The witness withdrew.)

Ms ELIZABETH WHITELOCK, Chief Executive Officer, Australian Information Industry Association, sworn and examined

The CHAIR: I welcome our next witness. Would you like to commence with a short opening statement?

ELIZABETH WHITELOCK: AIIA represents the breadth of the technology sector in Australia, from sole traders to multinationals. We have representation from telecommunications, from SAAS and cloud providers, to data centres, to SMEs who specialise in particular areas such as cyber, AI, and 53 per cent of our membership base are SMEs. The submission that we put forward was based on the feedback from those members, from large to small. We are not specialists in the data centre space, but obviously we have people who are, and we have people who use them, including Federal and State government. Our submission reflects the views of our members.

The Hon. SCOTT FARLOW: In your submission you argue that we should avoid the Singapore-style capacity controls and shouldn't rigidly link approvals to renewable generation build-out. We've also heard from the Business Council of Australia that that would not be feasible. I'm interested in your insights as to why you believe that is the case. What would be the detriment if we were to have those capacity controls linked to data centre build-out?

ELIZABETH WHITELOCK: We're playing in a space that is highly competitive. I was in Victoria last week at a data centre conference and Victoria were flying the flag, State against State.

The Hon. SCOTT FARLOW: Don't worry, New South Wales doesn't care, according to the Treasurer. They'll just come here anyway.

ELIZABETH WHITELOCK: We have internal competition within the country. Organisations are looking to where is there constraint and how do we avoid that? So power, land, skills and workforce—and, at a national level, that is one aspect that we have to consider as New South Welshmen. We need to think about what we can do to relieve those constraints, to be part of that competition. But, more broadly than that, we have international players also looking at where can we place our data centres? Where is safe? The great thing about Australia is that we have this safe and stable government world at State level and at Federal level, and that is an attraction. We are great at policy. All of those things are ticks in the boxes for us. But what it then boils down to is where do we go and how do we make sure that we have got access to all the things that we need access to?

Removing the constraints allows us to attract, at a local level, local data centres, but also international players and the hyperscalers into the market, which obviously is a boost to our economy and then a knock-on to productivity. If you look at Ireland, I think it was, there were policies put into place about access to power and what was expected back. So renewables giving back to the grid—those elements were formed in policy. I think, at the end of the day, if we want to be part of this digital infrastructure, which is going to enable our economy and enable productivity, then we have to figure out how do we create the right environment for organisations to want to come into Australia and specifically into New South Wales.

At a policy level, what do we do? I think that this plays to a broader piece, which is what is the overall plan here? What is the strategy? I've just come out of a meeting where the phrase used was, "It has to be digital strategy first, because that train has already left the station." We need to be part of that. What do we do to create that right environment where people are not constrained? Our policies about access to energy, what is it that we offset with the renewables that data centres are now producing and supporting the grid? Access to water—what might that look like? What are the innovations? Necessity is the mother of invention. Perhaps by setting the policy and the plan of attack, we start to drive those innovations by the data centres that are not already in place.

The Hon. SCOTT FARLOW: To that point, what is the plan? What is the strategy? Does New South Wales need a strategy?

ELIZABETH WHITELOCK: Yes, absolutely. You may already have elements in place. Ultimately, the fact that we're sitting here having the conversation about it would suggest that we probably need to fill some of those holes. We need to fill out what the constraints might be and how we can get around those constraints, not by backdoor deals, but by actually saying, "What's good for the economy? What's good for New South Wales? What's good for the citizens? How do we deliver better services to our citizens?" It has to be a broader strategy with digital as that foundational layer. Digital is critical. Digital is enabling. It is the equivalent of the poles and wires running down Macquarie Street. It's the equivalent of the telecommunications infrastructure that we have. It's the equivalent of the metro and the transport infrastructure. Without it, we are behind, so I don't think we can ignore it. It has to be part and parcel of everything that we do from a strategy.

The Hon. SCOTT FARLOW: Your submission highlighted that "thoughtful coordination can allow New South Wales to capture the economic and technological benefits of digital infrastructure while responsibly

managing environmental and community impacts". What are those specific coordination recommendations that you would have for us that we could put into a report and have as a recommendation to government? What should we be doing to coordinate to get the best economic benefits out of data centre development in New South Wales?

ELIZABETH WHITELOCK: Like any development—it doesn't matter whether it's a data centre or not—ultimately, there are going to be multiple stakeholders, and the stakeholders have to include the community themselves wherever that data centre is. In fact, probably more broadly than that, what does the community want—understanding the community's concerns and addressing those. Stakeholders are also private industry—the people who will build the data centres, the investors who sit behind it—and, obviously, government. Then within government there are the planning departments, environmental et cetera. It is a multi-stakeholder approach that's required. Leaving one of those constituents out of that conversation only ends up creating more angst, more concerns and, potentially, more delays. It has to include the community, and it has to think about all of the things that you've already highlighted around access to power and water and the immediate impacts on the local communities wherever they're built.

The CHAIR: We've talked a lot about what is critical, and we've talked a lot about how we get the so-called social licence or having community on board. Looking at what's happened, my take is that people generally agree that digital and technology is incredibly important and critical. Up until maybe the last year and a half, everyone was very on board with the pace that that was going and also in terms of thinking about technology as being almost a resourceless endeavour: It's something that happens; it's techy; it is kind of happening somewhere else. With the advent of AI and it becoming such a huge part of applications and everything else, and then this growth in hyperscale, really massive data centres, my impression is that people are seeing that as something different.

The question then is, from your perspective, is there a need to distinguish between what is actually critical—in terms of what we need to advance as a society to do all the things that we need to do in research and health and everything else and keep the planes flying and the banks operating—versus the question of whether we want to be taking huge amounts of water and energy and, potentially, putting back climate targets and all the rest of it for something that the public doesn't necessarily agree is in that same category—the cat videos, you know.

The Hon. SCOTT FARLOW: You really hate cat videos, don't you?

The CHAIR: I really don't like the cat videos. Anyway, one of your recommendations is around acknowledging data centres as essential infrastructure or critical infrastructure. Do you think that if we were to make a distinction between types of criticality that we could more easily bring the public on board for the types of things that actually are "critical, critical" as opposed to kind of "nice to have" or cat videos.

ELIZABETH WHITELOCK: It's a really good question and I think the cat video element has already gone. There's not much we can do about that. I think social media consumption, given most of the content probably comes out of the US and other places, is perhaps not an immediate concern here. The criticality element is absolutely spot-on. We need, as a group of citizens, to know that our data is safe, that it is not being off-shored anywhere. Those elements around health care, where is my data being held, for example; or banking and finance; government services et cetera—all of those things are absolutely critical because they are essential services.

If there is a distinction, is it a classification of essential services? My concern with flagging something by saying, "This is essential" versus not—how do we know that an SME is not out there building out some new innovation, potentially, that is consuming tokens somewhere along the line and one day actually sits within this essential service? How do you create the distinction? The SME sector being what it is in this country, and the backbone of our economy, employing 50 per cent of the workforce—the development and the talent that exists in this country is incredible. There are lots of people doing incredible stuff in health, as an example, or in aged care, that requires AI and data centres. Would we then remove them as "essential services" because it's potentially research and development, particularly given some of these things can take 10 years to come to fruition?

I would be worried about creating a very hard line in the sand that said, "This is essential", versus anything else. But I think part of that framing is to bring the community along. We have to physically understand what the implications are for the upscaled use of data centres. What is the real power and water consumption rate? I say that because I sat in this conference last week with people who are in data centres, people who are in property and people who are in investment and there was no one view that said, "This was going to get out of control with energy demands". It was all over the place. I think there's an absolute requirement, and this not our space because we are not data centre people, but I think we need to be really clear that when we say it's going to consume a lot of energy, what does that really mean in comparison with the rest of the economy and other industries and in comparison to us as end consumers?

I will come back to the phrase "Necessity is the mother of invention". I know from one of our members that they have done some incredibly innovative technology in recycling of water. Their use of water is really low and they use the same water again and again. There are elements that remove that concern. As a citizen, what I want to know is that our economy isn't going further behind, that we are going to see that our workforce actually has jobs, that we are productive, that our productivity levels go up. Technology—like it or lump it—underpins almost everything that we do. How do we change the narrative for the communities? I think it's about what is the reality of consumption but what are the benefits if we get this right.

The CHAIR: I think that's a really good answer because it's covered so many aspects of this. I was reading a Roy Morgan poll showing that 65 per cent of Australians think that AI causes more harm than it fixes. Part of the issue is around the lack of socialisation of benefits around technology. As a society, we've not done very well. When someone can take advantage of technology and they sack a whole bunch of people on the back of it, for instance, that is not met with any extra benefit for people as a whole.

ELIZABETH WHITELOCK: Of course, totally.

The CHAIR: I think there are issues around that. From a technical perspective—and there are two aspects, so maybe the question could be which do you do. Acknowledging that we can question how significant the impact will be, but there is a significant impact on energy and water resources, at least from what is proposed in the pipeline at the moment. Do we either seek to put limits on that use now, to encourage that innovation, without worrying what is actually happening within the data centres as much, or do we go the other way and say, "Let's get more transparency around which data centres are doing which things", so that then the community can more adequately weigh up whether they think it's worth the damage or the cost in terms of water and energy?

ELIZABETH WHITELOCK: There's the broader perspective, I believe, and that is what is the planning that we're doing around the grid. What is the planning that we're doing around water scarcity and access? Whatever we're doing up here with the energy grid and renewables feeds into the data centre itself. I don't think we can take one without the other. I think we need to understand what the broader picture is with the grid and its renewables et cetera. What is it that we want the data centres to do with their energy use? They already are deploying renewable capability and pushing it back into the grid. What do we want them to give back to the grid so that we balance it? Perhaps that's more about a policy framework around what can be done to mitigate the perceived issues around energy and water consumption. They all go hand in hand. I don't think we can simply remove data centres and say, "This is a problem." I think we have to think about it in a broader context of everything that we have to consider from our economy and the supply and demands of energy and water generally.

The Hon. JACQUI MUNRO: Thank you for coming and for the submission. I have a question about the essential digital infrastructure as well. Do you think that we use the Federal Security of Critical Infrastructure Act as our guideline for what we consider in the State critical infrastructure? Is that a useful framework?

ELIZABETH WHITELOCK: I don't know it in detail, I have to say. But I would say, observationally, we probably don't consider that in light of what is regarded as critical infrastructure at a legislative level, which is why I think we also need to take a step back and say, "This is new." We've been using cloud providers for a very long time in technology and in government and, at some point, who knew that we were going to end up in a place and a time where the world has shifted so dramatically? I think it's time if it's not considered that it probably should be considered because, if we're going to continue to be part of a global economy—and I suspect that we are—then digital infrastructure like data centres has to be considered.

The Hon. JACQUI MUNRO: Do you see that there is a role for data centres to play in our geopolitical strength and in relationships with other countries, and how do you see that?

ELIZABETH WHITELOCK: Good question. If there's one thing that the last few years has taught us, it's that uncertainty is like taxes and death. It seems to be, yes, it's absolutely happening, from COVID to where we are today with the Middle East. And what did we learn through this process and what are we learning today? That sovereignty and domestic capability are really, really important, and relying on others for some supply is clearly an issue. How do we create that domestic capability, that sovereign strength, which is not just about the infrastructure itself; it's about a skilled workforce. It's about ensuring that our workforce has all the right training and access to learning that they need to do their jobs, because ultimately—and this is a very personal opinion—the A in AI is "augmented". It is about bringing a set of tools to individuals so that they can do something—whether it's maybe manual and repetitive—faster, more efficiently, to allow them to do the things that are human, with all of the values that a human brings: empathy and sympathy et cetera.

The Hon. MARK BUTTIGIEG: Can I take you up on that point a little bit because I find this a very interesting argument. I get the sovereign capability argument for tangible things like the production of aluminium or manufacturing a local base so that we have goods and services which are tangible that could be otherwise cut

off, like oil or whatever. When it comes to data, I struggle a little bit with this kind of binary argument or what seems to be a special argument for data centres. When you think about it, we're already transacting billions of pieces of information across the world, be it internet banking or whatever we're doing, so why is it so important—because you can imagine a world where data centres are multiplying hand over fist all across the globe.

We're interconnecting with submarine cables. Why is it so critical that we have to have—and I'm just posing this as a bit of a challenge to the assumption that we necessarily have to have them because I'm interested in it. Why is it so important that we have to get ahead of everyone and have more data centres than everyone else? Isn't it more about our intellectual property and developing the AI models, for example, that's going to add the value rather than just having data centres? It would be interesting to know what your perspective is on that.

ELIZABETH WHITELOCK: First of all, certainly we haven't suggested that we should be ahead of everyone else in the number of data centres that we have. If we wish to play in the global economy, we need to have infrastructure to allow us to play. To your point on AI and building models, where are we going to build models? Where are we going to test them? How are we going to scale them? All of that requires some form of infrastructure. You can build the data centre internally, and you can have all of those servers running all of your AI and training your models ad nauseum, or it sits in a data centre that is independent. Either way, if we're going to be in the AI economy, then we need infrastructure to be able to build those models, train them and continue to evolve them.

The Hon. MARK BUTTIGIEG: This is my point. For example, let's have this bizarre experiment where we didn't have any data centres. They're all domiciled in Singapore, and we've got submarine cables. What's stopping us doing that?

ELIZABETH WHITELOCK: There's not. You're absolutely right. There is no reason not to. In fact, you'll probably find that with some of the hyperscalers we've had data sitting in the US and other places around the globe. The question then is, to something that the Chair has kind of touched on, what does the community want? If you look at the breaches that we've had in the last few years where data has been sold on the black market on the internet, people are lacking in trust. So, ultimately, what do we do to ensure that we have the capacity to scale, the security that's required that involves maybe encryption and so on?

What do we do to do that so that our citizens back the idea that technology is an underpinning platform that we need to continue to evolve? Yes, we can offshore it, for sure. The question is where would you then offshore it and what levels of regulation would we set around that? What policies would we want to put in place? What agreements would we want to put in place with those foreign jurisdictions—all of those things. Do we provide backdoors into Singapore so that we can access our citizens' data? They are all possibilities but, ultimately, as an Australian citizen, I'd like to think that my data is secure here, being used by NSW Health, by the ATO, by Services NSW. I would like to think that it's safe and secure and I don't have to worry about it.

The Hon. MARK BUTTIGIEG: That's an interesting point. My question is, is that more an emotional response about visualising the location of data rather than a function of cybersecurity?

The Hon. JACQUI MUNRO: You've got to make sure someone's not going to cut the cable and stop us from accessing any data.

The Hon. EMILY SUVAAL: Is that a question to the witness, or a discussion?

The Hon. MARK BUTTIGIEG: I'm happy to have a free-flowing debate.

The Hon. EMILY SUVAAL: It's probably not appropriate to do it now.

The Hon. MARK BUTTIGIEG: But to that point, if that were to happen now, we'd completely lose the world financial system, wouldn't we?

ELIZABETH WHITELOCK: Unfortunately we don't live in a time and place where everyone is a good actor, so resilience at an Australian level, sovereignty of our data, I don't think anybody would question, personally. Certainly within our membership and within our ecosystem, I don't think anyone questions the sovereignty of the data, the security of our data and having that here. I'll come back to a point I already made, and that is that we have this benefit that we have a government that is stable and that is well regarded on the world stage, which means at a local level we can build our own infrastructure for our own purposes. But it also means that we have an opportunity to attract international investment and international players into the market because of the stability. They know where the constraints are, they know that the skilled workforce is there, and they know that they've got a government who is stable.

The CHAIR: Unfortunately that does bring us to time. Thank you so much. That's been really interesting. There will be questions—I know I've already got some supplementary questions—and to the extent any questions have been taken on notice, the Committee secretariat will be in touch. That concludes our time with you.

(The witness withdrew.)

(Luncheon adjournment)

Mr GLENN HUGHES, General Manager, Sydmech, affirmed and examined

The CHAIR: Welcome back. Would you like to start with a short opening statement?

GLENN HUGHES: I'm okay.

The Hon. MARK BUTTIGIEG: Thanks for appearing, Mr Hughes. You might just want to give us a quick outline of what you're involved in, in terms of the perspective you bring to the inquiry, just for the Committee members.

GLENN HUGHES: I've provided a bit of insight into the mechanical contracting component to the Plumbers Union submission. I've been working and I currently work for a mechanical contracting business. However, that business also provides direct labour to the industry and has predominantly been working in the last seven years within the data centre component of the industry and sector.

The Hon. MARK BUTTIGIEG: So a lot of your demand is coming from seven years worth of data centres?

GLENN HUGHES: Yes, it has.

The Hon. MARK BUTTIGIEG: Do you want to give us your perspectives on where you see potential problems and opportunities with this growth industry? We've had a lot of evidence to date regarding the proliferation, the demand for electricity and water based on modelling, which is a bit nebulous to get a hold of because of connection rates versus actual real demand. I know there are issues with skill shortages and all sorts of things. Just feel free to give us a perspective, from someone who actually is involved in the day-to-day building of these things, on what you think are the main obstacles to us getting a benefit for the taxpayers of New South Wales.

GLENN HUGHES: If I could start with saying the proposed data centres that have been discussed in our industry are going to be good for our market in New South Wales if we can bring them here. The last couple of years have been rather repressed in the construction industry, and I think it definitely would bring some vitality back into the market. In respect to concerns, and if I move away from the labour, which I'm sure you've heard a lot of—

The Hon. MARK BUTTIGIEG: We did hear evidence from the Plumbers Union on the issue of skill shortages and training and everything, but feel free to add to that if you want.

GLENN HUGHES: The New South Wales market in probably the last 15 years has been predominantly driven around, at the high end, hospital-based work with the sheer amount of hospitals that have been done. There are a lot of general tradesmen that are associated with normal air-conditioning systems built around buildings and healthcare facilities. I don't think there's really a shortage there. But the data centre market brings a different perspective to the mechanical services. The level of mechanical pipework and plumbing—and I'm talking just from a mechanical perspective—is near on double that of, say, a normal healthcare facility such as one of these larger hospitals.

The Hon. MARK BUTTIGIEG: That's really interesting. If you took a hospital like Nepean, your average data centre would be, what, double the requirement for mechanical services—air conditioning, cooling and all those sorts of things?

GLENN HUGHES: Put aside the design part and capacity. With regard to labour and the quantum percentage of the project, it is definitely at a higher level. You're talking about more plumbers on site or more plumbers in factories producing pipe work for much larger, expansive systems compared with your average project. I've compared that with health care because health care, such as Nepean—and I was involved in the design of Nepean. Nepean has got a pretty expensive chilled water system, but your average data centre has a lot more pipe, a lot more water, a lot more chilled water plant serving those facilities. The percentage of, say, ductwork versus pipe work has shifted. It has almost inverted.

The Hon. MARK BUTTIGIEG: Is that more labour intensive and more specialised work?

GLENN HUGHES: The trade requirements of mechanical pipework are at a higher level in our market at the moment. They're done by licensed tradesmen, whereas for sheet metal there's a transition from non-tradesmen to tradesmen now with the proposed changes in legislation for licensing. If I context it right, with the shift in quantum of work, the numbers of qualified tradesmen that can weld pipe and deliver the installation of pipe to the market—

The Hon. MARK BUTTIGIEG: Tradespeople.

GLENN HUGHES: Tradespeople, excuse me. It is still set at the moment based on a traditional proportion of mechanical plumbing. With that proportion changing, the number of tradespeople is going to need to increase and, therefore, the training, the opportunity, the attraction of even apprentices to the market needs some focus because we need that skill set to deliver the quantum that we have heard in data centres is going to be brought to the market.

The Hon. MARK BUTTIGIEG: Just so the Committee understands, mechanical services is a subset of the general plumbing trade but it's different to what you would traditionally understand a plumber to be—in other words, someone who comes out and plumbs your shower and bathroom and toilet drainage or whatever. This is, if you take this building, the cooling systems, isn't it? It's the high-end cooling systems and all the associated pipework and the chillers. That doesn't necessarily get trained in the TAFE sector, does it, for historical reasons?

GLENN HUGHES: Historically, many years ago, mechanical plumbing was an optional component that could be trained in at TAFE, when the tradespeople were doing their trade and undertaking it. My understanding is that has backed off quite dramatically. Generally, my understanding is that they teach and train for sanitary plumbing. If I term it that way, it might be easier.

The Hon. MARK BUTTIGIEG: Which is everyone's traditional perception of what plumbing is.

GLENN HUGHES: Where our plumbing for mechanical plumbing is chilled water pipe work, it does involve some copper, but different. It's steel pipe. It's large-bore pipe. They are closed and open water systems, with pumps associated with them, connected to plant that heat and cool the water within. It is the facilitator of the media that we transfer heat with.

The Hon. MARK BUTTIGIEG: We also heard on evidence from the plumbing union about the requirement for adequate fire protection as part of that. Do you get involved in that too?

GLENN HUGHES: Fire protection for mechanical services is more associated with penetrations and breaching fire-rated walls and structures primarily. If we went backwards to other components—and it's not so much in data centres—we also provide services to comply with the NCC, the national building code, relevant to smoke management within a building.

The Hon. MARK BUTTIGIEG: This demand for trained mechanical services people, do you see that as being potentially a barrier to the speed with which we can build these data centres?

GLENN HUGHES: It's my belief it's a risk to delivering the scale that has been discussed in the market. It will certainly depend on how they phase those works into the market and over what period of time as well.

The Hon. MARK BUTTIGIEG: We heard evidence from Mr Tsiakoulas from the plumbing union. His view was that what would happen is the market would bid up wages, salary. In other words, because of the proliferation of data centres, you'll have a drain of skilled labour from the general construction sector over to data centres, because they'll be working overtime. Then that could, in turn, have an impact on housing construction, for example. Is that a valid concern?

GLENN HUGHES: In my opinion, yes, it is. It is definitely a valid concern. If we've got a skills shortage in a particular trade, in trying to meet deadlines and phasing, certainly the construction teams will work longer hours and they'll work weekends and the like. Depending on the scale of jobs, it could actually shrink availability of labour to other parts of the industry in the building sector. For instance, there are several hospitals that have been proposed recently and that are forecast to be done. It may actually strip labour from those infrastructure-based jobs as well.

The Hon. JACQUI MUNRO: We heard evidence earlier from Dr Sue Keay, who suggested that, given the increasing proliferation of big construction projects from the Brisbane Olympics to data centres to housing, robotics could be used more effectively to assist in construction. Although she came here in her capacity as the Director of the UNSW AI Institute, she is also the chair of Robotics Australia Group, so she comes to it with some exceptional expertise.

GLENN HUGHES: It's not my speciality or expertise with regard to robotics, but we are aware of certain trades within the building industry more so which are actively adopting robotics. There are robotic bricklaying machines and the like. For our profession, I guess the limitation at the moment is material handling. You can get robotic welders. They're not typical in our industry. I believe it would be more associated with other trades more than ours.

The Hon. JACQUI MUNRO: It would be more about freeing up labour from those other types of construction to get them to a more specialised—

GLENN HUGHES: That's a good point and a good statement. Yes, it would potentially shift a balance of labour that could be retrained—and I would be very surprised if the plumbers union hadn't actually discussed this—into other fields of the industry. I can't comment on how well that would work. I believe the plumbers union would have a better view on that than I would.

The Hon. MARK BUTTIGIEG: From an industry perspective—someone who's into this up to their neck, by the sounds of it—what's the key take-out? Is it that the Government needs to start thinking about investing in more bespoke training to facilitate the building of these centres?

GLENN HUGHES: My personal view with regard to current training is that training has been in a decline and potentially, in some parts of our industry, inadequate for a fair while now. It's one of the reasons why the sheet metal trade has all but disappeared in our part of the industry. In retraining staff, I would believe that a bespoke, specialised training regime would need to be applied if we were rapidly trying to retrain staff to respond to the building and trades load that we would be expecting.

The Hon. JACQUI MUNRO: My questions are on a similar line. How closely do you work with TAFE or other training centres to try to assist them in building a curriculum that assists you in your work?

GLENN HUGHES: We send apprentices to TAFE continuously, but we also do in-house training as well. A lot of the general hand skills come from the workplace, with tradespeople training apprentices accordingly and progressively exposing them to new environments and new skills over time. In my opinion, the answer to your question is rather mixed, and it would depend on the specific part of the trade. I can comment, because my son is actually doing an apprenticeship at the moment, that I believe there's been a pretty big decline in TAFE services offered to the refrigeration and air-conditioning component of the industry. With regard to plumbing, I can't really comment, I'm sorry.

The Hon. JACQUI MUNRO: No, that's helpful. On apprentices, are you finding that when you're able to get them into these quite specialised roles, they are demonstrating a higher rate of retention?

GLENN HUGHES: We've been lucky as a business. Our apprentice retention has been pretty reasonable. However, recently—in the last 12 months—we've doubled the amount of apprentices we've been bringing in to the business. In the last four months I've employed three new apprentices to our industry.

The Hon. JACQUI MUNRO: Awesome.

GLENN HUGHES: The difficulty that we find is in attracting apprentices to the market. I'm not sure whether it's a demographic change but, certainly, attracting young apprentices and young people into our market's pretty hard. There are some agencies out there that are encouraging it. We've been participating in some of those recently on jobs where the social procurement agencies are looking at young, unskilled employees, bringing them into the market and giving them a little bit of training for all sorts of roles across the construction industry. We are currently trialling two employees from that system. The initial feedback I give you is it's pretty good. We've got some people that have had long-term unemployment who we've brought in for training as an apprentice, in this case, in sheet metal. We've also got a new apprentice that's never had a job who is in his twenties, in this case, which I think is a big thing.

The Hon. JACQUI MUNRO: Absolutely. I think that's fabulous. You said you were lucky to have had a pretty good retention rate with the apprentices that have come through. Are there things that you can point to that have assisted in that retention? I suspect it's not just luck.

GLENN HUGHES: No, it's not just luck. In the businesses that I've worked for, we've concentrated on, in our trade labour, making sure we get as many of the qualified tradespeople as we can within our field, be it mechanical plumbing or sheet metal. If I cast my mind back 15 years ago, the greater majority of sheetmetal workers that were actively working on jobs that I was project managing at that point in time were technically skilled labourers; they weren't trade-qualified personnel. Generally, in my opinion—and it's purely my opinion—I don't believe that in the last 20 years the industry has invested enough into bringing new apprentices, new trades or even bringing engineers into our industry and training them to do the job within our field of the industry, be it engineering, mechanical plumbing or sheet metal.

The Hon. JACQUI MUNRO: Do you think that this actually does present an opportunity, in terms of the data centre boom, to bring in more apprentices, to offer people more jobs and to actually have this pipeline of work being hopefully more stable, to allow workers to be skilled up and trained up more than they might have been in the past?

GLENN HUGHES: I believe it's an unbelievable opportunity to bring and train new tradespeople into our industry. The concern I would have—and this is a personal concern—is making sure the ratio of currently trained tradespeople to apprentices is correct. I say this in simple terms: If the percentage of apprentices is way

higher than the tradespeople, the level of exposure the tradespeople can give to each individual apprentice is going to lower. Potentially, my concern is that without supplementary support and training, the quality of tradesperson delivered at the end of the apprenticeship may drop.

The Hon. JACQUI MUNRO: Fair point, of course. Do you have a ratio that you generally use for your own business?

GLENN HUGHES: I have a view that you shouldn't exceed more than 20 per cent in apprentices. In reality, it's often hard to get to that 20 per cent range in attracting appropriate people to take on new trade work in our industry.

The Hon. MARK BUTTIGIEG: My colleague the Hon. Jacqui Munro's point is well made. In your evidence, what you're telling us is that some of the jobs that should have been proper trades and got proper training from the vocational education sector have fallen away. As a result, you get these semiskilled people coming in and learning informally on the job. With the advent of this demand for data centres, it's an opportunity to both increase the supply of labour by training and also clean up the training regime as a by-product. You get this virtuous circle if we do it properly. Is that the main message for the Government?

GLENN HUGHES: I think that's a reasonable take on it. I have a view, and it's probably very aligned with the union, that it needs to also be driven by a licence requirement to ensure that we take it from where we are today—and in a lot of cases, it's more associated with the sheetmetal component of our industry, and I'd be surprised if you hadn't already heard that—and bringing that to a point that when we upskill the staff. There are a lot of staff in our industry that are at a point where they should be truly qualified tradespeople because they've done it for so long.

The Hon. MARK BUTTIGIEG: They've basically been doing it in practice, and then they need it recognised.

GLENN HUGHES: That's right.

The Hon. MARK BUTTIGIEG: I'm conscious of the time, Mr Hughes. In terms of data centre density, if you like, on the materials, is sheetmetal ducting work—I think you said this in your earlier evidence—quite a big proportion of data centres?

GLENN HUGHES: In a normal job, like in a hospital, it's a greater proportion than mechanical plumbing.

The Hon. MARK BUTTIGIEG: As a proportion of the job?

GLENN HUGHES: In a data centre, it's typically inverted, and it's a lower proportion of the job than mechanical plumbers. Just to confirm, that poses my concern as to whether or not we can support enough tradespeople into those data centre jobs, given that on typical jobs that we've been doing—predominantly in commercial buildings and then health—over the last, say, 20 years, there's a set number of mechanical plumbing tradespeople in the market. The training is, in my understanding, a little different. It's a supplement on top of your sanitary plumbing trade.

The Hon. MARK BUTTIGIEG: When you do the mechanical services part of your plumbing—if you choose to do that—you would do both the pipework and the sheet metal work?

GLENN HUGHES: I can't comment on that. My trades that I have are not in mechanical plumbing. It's been refrigeration.

The Hon. MARK BUTTIGIEG: I understand.

The CHAIR: There's been some talk of whether, given we've got—and this happens whenever we get a new industry that's booming. We always look back and say, "Wouldn't it have been nice if we'd have mandated as many local content requirements as possible?" Would you support us mandating a certain percentage of local content requirements? I understand there are some components in a data centre build that you cannot get locally, but there must be some that you can. Would you support that?

GLENN HUGHES: Yes. Just for reference, we have been, in the industry, mandating at a DA level local participation, female participation and long-term unemployment in government jobs such as health care. Those requirements have been actively around for probably the last 10 years or thereabouts, is my recollection. If I were to add to the comment of mandating, I do believe that there needs to be a view taken on energy consumption and water consumption, and ensuring efficiency in designs to lessen the burden on the infrastructure supporting these facilities and the infrastructure that also supports the community at the moment. In saying that, I believe we should, like with health care and commercial buildings, have minimum energy performance and water

performance criteria applied to these jobs, so that burden can actually be managed as it's applied to the infrastructure. That is my personal opinion. I'm sorry. That's beyond what you asked.

The CHAIR: It's really good, though. Can I just clarify? When you said that those local participation requirements and other things had been mandated in the last 10 years, did you say at the development consent level, at the DA level?

GLENN HUGHES: I believe they've been applied at the development consent level. If I use the previous reference to the healthcare job of Nepean, Nepean had a local content requirement relevant to local participants being employed on the job—under 25-year-old employees, apprentices, female employees, Indigenous participation. These are common to what Health Infrastructure has been applying to the projects that it has been doing. My understanding is part of that is coming from a DA and local government, but I also believe it's also potentially coming from Health Infrastructure because it seems to be consistent whether or not you're working in Sydney or, say, Griffith, Bega or Byron. It seems to be applied consistently across those types of jobs.

The CHAIR: That's very useful. Thank you. I may have some supplementary questions. To the extent that there are any further questions, the Committee secretariat will be in touch. That concludes our time with you.

(The witness withdrew.)

Ms EMMA BACON, Executive Director, Sweltering Cities, affirmed and examined

Ms SANAA SHAH, Senior Community Campaigner, Sweltering Cities, affirmed and examined

The CHAIR: I welcome our final panel of witnesses for today's hearing. Do you want to commence with a short opening statement?

SANAA SHAH: Sweltering Cities works with communities impacted by extreme heat to advocate for more liveable, equitable and sustainable homes and cities. Our organisation was founded in Western Sydney in response to the region's disproportionate heat vulnerability. Heatwaves are already Australia's deadliest environmental disaster, and Western Sydney experiences these impacts most acutely through higher temperatures, rising heat-related illness and soaring energy bills. A growing concern is the expansion of hyperscale data centres in some of Sydney's hottest suburbs, including Marsden Park, Eastern Creek and Kemps Creek, where a one-gigawatt project has been proposed.

In our submission, we have integrated 32 comments from supporters living near proposed or recently approved hyperscale data centres. These highlight concern that Western Sydney is increasingly being positioned as a data centre hub without adequate consideration of local impacts, including urban heat, noise, water use and pressure on the energy grid. I'd also like to read out one of these comments that were submitted to us that we have included within our submission. This comes from someone who lives in Marsden Park, just a minute down from the hyperscale data centre that has been proposed there. She says:

No one in the community I've lived in my entire 29 years of my life wants this data centre here. We have already had this area destroyed due to poorly planned housing developments, please rethink this decision- data centres are 100% unnecessary to life- we've lived our entire lives happily without them. But fresh air, water & climate is something no one can live without.

Given all this, there's a serious risk that cumulative impacts will deepen existing socio-economic and environmental inequities in Western Sydney. We also want to draw attention to the Western Sydney Local Health District's concerns against the CDC Marsden Park proposal. The Western Sydney Local Health District warned that neighbouring communities already experience high rates of chronic illness, and that additional exposure to heat, noise and air pollution from that data centre could worsen health outcomes and further strain local health services.

Our submission is also informed by joint work with partner organisations which have made their own submissions to this inquiry, including the Stop the Bill Shock coalition, which highlights the risk of households absorbing rising energy costs, and the Tomorrow Movement, which amplifies the voices of young people in Western Sydney. We are calling for a pause on further data centre approvals until there is clear evidence that their social, environmental and economic impacts are being properly assessed and managed through strengthened safeguards. These should include heat impact assessments and protections to ensure rising infrastructure demand does not translate into higher energy costs for consumers. More broadly, we strongly believe that planning priorities in New South Wales should direct investment towards social and community infrastructure rather than the continued approval of hyperscale data centres with limited local benefit.

The CHAIR: Thank you very much both for appearing but also for your excellent submission. I thoroughly enjoyed reading the comments, in particular, from community. They were really incredibly insightful and very useful. I understand that the Western Sydney Local Health District raised formal objections about cumulative health risks at Marsden Park with increased ED presentations, mortality risk and chronic disease impacts in Bidwill, Shalvey and Hassall Grove. Your submission says that those concerns were dismissed. Would you support a requirement that any future data centre SSD application in Western Sydney should first obtain a formal health impact assessment from the relevant local health district before approval?

SANAA SHAH: Yes, we'd certainly agree, especially for a key stakeholder like the Western Sydney Local Health District.

The CHAIR: We've spoken to a few different community groups who are facing the prospect of having a data centre quite close by to their homes, and health hasn't been one that has come out very strongly. We have heard a bit about concerns about air pollution and noise. What is your experience of where people are concerned and where the health risks would be, particularly when it comes to heat? Because we've not really heard much about that.

SANAA SHAH: Yes, in the conversations we're having—both through the comments that we integrated within the submission but also other ongoing campaign work that we're doing—is the fact that these data centres could impose additional heat stress on local communities, which are already so heat vulnerable, and there are associated health risks that would come with that. The specificities of the Western Sydney Local Health District's comments on ED presentations are not something we've heard too much of either. But I think it's definitely one

of those underlying concerns that also surround the other impacts, like impacts on air quality, water quality and noise pollution.

EMMA BACON: Because heatwaves are the biggest killer of any environmental disaster in Australia and because Western Sydney faces a higher burden of disease when it comes to heat, it's an area where we need to have a higher standard of concern as well; the communities we're talking about are already facing a much higher rate of death, illness and disability due to heat. This isn't just a question of heat-related illnesses during a heatwave, even though those are really severe outcomes; there are also significant mental health impacts of heat.

Being exposed to hot night-time temperatures—which is more likely if you live in an urban heat island like Marsden Park where the environment doesn't cool down overnight; and we're talking about 24-hour operation at data centres, potentially that will also have hotter overnight impacts—means that people aren't sleeping. There are increasing rates of anxiety and depression; youth suicidality goes up with temperature as well. Those are all impacts that will be felt in our health system. We also know that domestic and family violence goes up in the heat. Those are all really significant health impacts being felt due to heat, so they all need to be prioritised when it comes to the concern about what are we investing in in our communities.

The CHAIR: Sometimes I speak to people about having a black roof or not having enough surfaces that absorb heat and all the rest of it. Can you describe, in as simple terms as you can, how having something as hot as a data centre in a suburb makes a difference beyond the data centre area and how that is felt?

SANAA SHAH: As simple as possible—the urban heat island impact is created when you have an abundance of dark surfaces, dark materials that all absorb heat and radiate it. This is something that can be felt in the local area. It's definitely something that is experienced in Western Sydney where you've had greenfield development on formerly green spaces or farmland that's all been paved over with dense housing. In this case, with these massive industrial data centres, that is—I know there have not been super conclusive studies internationally on the additional heat load that can be brought about by these data centres, but there is definitely evidence that suggests there will be up to a 10-degree or 15-degree impact from these data centres being constructed.

EMMA BACON: It's also important to note the context of these environments where they're already facing a high degree of the urban heat island effect. It can be 10 or even 15 degrees higher than surrounding rural areas or coastal suburbs. In Marsden Park, it's not very far from—in January 2020 they recorded 48.9 degrees, the hottest temperature ever recorded in Sydney. The Western Sydney University, Blacktown council and Penrith council have also done local heat benchmarking studies where they've taken local temperature readings and found temperatures over 50 degrees already occurring in those areas. So when we're thinking about it, that's already extremely dangerous heat for people. Data centres, we know, are contributing to the problem; they're certainly not solving it.

The CHAIR: The obvious next step then is that if you're living near a data centre and the data centre is consuming huge amounts of energy, people in the surrounding neighbourhoods are presumably needing to put their air conditioning on more and consume more energy. It seems like we're in this doom loop.

The Hon. JACQUI MUNRO: Can I just ask about that? I totally understand where you're coming from in terms of Western Sydney in particular, and I love your work. I understand that you have a big building as a data centre that takes away from other, say, green space. But I'm just trying to clarify, is there an ambient heat effect from the data centre buildings? Does that form part of the concern?

EMMA BACON: In our submission, we identify a study that estimates that data centre electricity usage is converted into low-grade waste heat, so ambient heat, that can be 30 or 40 degrees Celsius. This is the same experience as walking around the edge of a shopping centre where you have these really large air-conditioning units that are facing outwards in order to maintain temperature on the inside.

The Hon. JACQUI MUNRO: Sorry, go on.

The CHAIR: No, that's good because I think that's what I was trying to get at in a roundabout way.

The Hon. JACQUI MUNRO: Ambient.

The CHAIR: That's right—ambient. Your submission describes the CDC Marsden Park approval as occurring within 12 months under the State significant development pathway, with community objections set aside. In your community engagement work, what were residents told, if anything, about the opportunity to have their concerns heard? At what point in the process were they even aware that the development was being assessed?

SANAA SHAH: We've heard that there was almost no community consultation, at least from folks who live very close to Marsden Park. We have someone who lives a minute down the road, but in other community

engagement stuff we've done there are also people who live very close to the site. They say that they didn't hear anything about it. They felt like they were quite tapped in to environmentalism but just weren't aware that this massive data centre was being constructed, and so that really calls into question what sort of community consultation processes were in place. I don't think that is clear for us at the moment.

The CHAIR: You call for genuine community endorsement throughout the planning process and long-term operation. What do you imagine that looks like in practice?

SANAA SHAH: For example, if there were heat mitigation plans, we'd want that to be done in a way that really emphasises local benefit and circularity. There are some international examples where ambient heat or the residual heat of data centres is being used to heat neighbouring buildings, like residential buildings. Whatever sorts of plans there are to mitigate these environmental impacts—

The CHAIR: Is that the waste heat recovery plan stuff?

SANAA SHAH: I think so.

The CHAIR: Is that what that's called?

SANAA SHAH: Yes.

The CHAIR: Some people have told us that in other jurisdictions you get things like a swimming pool being heated by a data centre, so capturing it rather than allowing it to leak out.

EMMA BACON: The two things, in our experience, on lots of different types of infrastructure that really are needed for community faith are transparency and benefit. People really need to feel like they understand the process—why are things happening; are there proper environmental assessments; how is the energy and water situation working—and that they're hearing that from reliable sources, not from social media and not from each other.

Obviously there are big concerns that are really legitimate. They need proper communication from both the companies and government about what's happening and to feel like they're legitimately being listened to from a very early stage, not at the very end. But also that if in their community there's going to be this hyperscale data centre being built, what benefit is there to them? At the moment, people feel like they're losing out to private companies, whereas what can be done to say there needs to be public infrastructure, there needs to be energy affordability and there needs to be guarantees that the communities aren't losing? I think at the moment, we have neither the transparency nor the benefit.

SANAA SHAH: In response to the Kemps Creek data centre, the Mamre Road one, there were some submissions made by the neighbouring schools. I went to that site a few days ago, and it's literally the big proposed data centre—they've already fenced it off a little bit.

Then, opposite the road, there's an Anglican school, there's a K to 12 school, and some of those schools made submissions opposing that project. They were saying, "We understand the economic benefits that this could bring to Western Sydney, but there was just no consultation." Things like construction impacts, also the potential air quality, health, noise pollution—because we know that those data centres emit that humming sound—none of that has been explained to the school community, and that's deeply concerning. We think, once again, this is literally a neighbouring—you know, there are a couple of schools there. There are young people there. There's no reason why that shouldn't be prioritised very early on in the planning approval process.

The Hon. JACQUI MUNRO: Definitely.

The CHAIR: We've heard some community groups talk about how ugly the data centres are, that they are just a big box. It strikes me as strange that there have been no efforts to try to improve not just the visual amenity but also the design function from the perspective of something like an urban cooling effect. Are there things that we could mandate, perhaps, for a data centre design that would allow it to contribute, or at least offset, in terms of urban cooling?

SANAA SHAH: Lots of trees.

EMMA BACON: The Greater Western Sydney heat plan that was launched by Minister Penny Sharpe in late 2024 is part of a comprehensive plan, consulted with many government agencies, led by the Western Sydney Regional Organisation of Councils and Resilient Sydney, which Sweltering Cities also sits on. One of the key recommendations of that is blue and green infrastructure investment across Western Sydney and across the entire city. Blue and green infrastructure is what cools down our suburbs. It's trees, but it's also the use of water in areas to make sure that we're reducing the urban heat island effect. I think that if the Committee would like any guidance on what that should look like, there are extensive resources we could send as well to say that there are—like, this

government and its agencies actually are some of the best in the country when it comes to understanding and planning what that blue and green infrastructure should look like. You just need to match those standards with compelling data centres to comply with them, through regulation, I think.

The CHAIR: Well said.

The Hon. JACQUI MUNRO: We've heard from a fair few witnesses that there can be really positive energy cost reductions, that grid stability can be improved, through data centres. I guess you sort of said it then. If we can demonstrate community benefit and if there is proper planning and process, do you think that there will be confidence in a data centre rollout? When you're speaking to people, are they saying, "We don't want data centres at all"? For example, one of those submissions said that data centres are 100 per cent unnecessary to life. While I understand where that person is coming from, we do actually use data centres for many, many things that underpin our quality of life. Do you think that there is a way in to have this conversation with people who have become disaffected by what's already happened?

EMMA BACON: I think, in a scenario where communities feel like there is, firstly, a comprehensive government plan for where they should be, not just in their community but statewide, I know there have been suggestions from—I was recently at the Energy Consumers Australia conference. That had a real focus on AI and data centres. One of the executives from Energy Australia, I believe, suggested that perhaps they could be co-located with large renewable energy, as a way to connect energy supply with energy demand—things like that.

So I think that if people could see that, firstly, there was a statewide plan for where they should be and why; secondly, that there was transparency about the environmental and the health costs and how they were being addressed; and thirdly, if there was clear benefit that was tangible, whether that's community investment in greening, in local infrastructure—we know that Marsden Park is still waiting for a library and a community centre and a pool, but they're getting a data centre. If those things were happening, if it was also saying data centres have to supply their renewable energy and any excess goes to the community, which they get for free—if we listed all those things were happening, then I imagine there'd be a lot fewer people who would have the concerns in that future scenario.

SANAA SHAH: That's also what's reflected in the conversations we're having. There'll always be a variety of views, from people who just don't want these data centres at all, to folks who are like, "We want them but we just need to do it in a more sustainable way." We need to welcome that variety of view as well. I think the core of the issue—and why we're here today—it's just there is no transparency, accountability and there's no community endorsement, so I think we need to be working on those.

The Hon. JACQUI MUNRO: Just to be clear, it sounds like you think a New South Wales data centre strategy would be a good idea.

EMMA BACON: Sure. I think major infrastructure should have a plan.

The CHAIR: That's radical.

The Hon. JACQUI MUNRO: I know. Who would have thought? It doesn't sound so complicated when you put it out like that.

The Hon. MARK BUTTIGIEG: For a party who doesn't believe in planning, they're big on planning. Can I ask you, some of the evidence we've heard is this advantage of going to these closed loop, water cooling systems. Have you got any insights into what ambient heating effect those systems have as opposed to the traditional—traditional in this market—but just getting water and pumping it through from the general water system? They're saying that you can have these closed loop systems and use refrigerants, I think, to cool it down rather than rather than draining Sydney Water resources. I just wonder how that affects the ambient heat and all those sorts of things you've been talking about.

EMMA BACON: We don't have any specific information on the temperature impact of those different cooling mechanisms. All of that type of refrigeration you're talking about does create ambient heat. We haven't heard of any solutions that don't have that. I also note that, as part of our submission, there is diesel generator backups for these centres that we know also create a lot of heat. We don't have specific information on those temperature differences, I'm sorry.

The Hon. MARK BUTTIGIEG: Is there anyone who you know of that's doing this in a way that attenuates all those negative effects you've put on evidence here today? Is there someone we could look to as an example of doing it really well?

EMMA BACON: No, I don't have any in particular. I do think that it's important that any examples that we use for something that is positive is being tested in similar conditions. We know that there's been an ongoing

problem in parts of Western Sydney, in Australia, where you have materials that are being tested in, say, a European context where it doesn't get as hot as it does here. You have play equipment, for example, that is tested for European weather but actually now reaches 80, 90 degrees in children's play equipment because it's just a different context. The same happens with some chemicals that are not tested in the heat that we have and, therefore, are destabilised and become more dangerous. If we are finding examples of places where they are doing what you're talking about—like having less environmental impact—but it's also 50 degrees where they are, then that would be an appropriate example for us to adapt to. But we shouldn't be using examples from places that don't have the extreme temperatures of Western Sydney.

The Hon. MARK BUTTIGIEG: Sounds like we need a trip to the Mediterranean.

EMMA BACON: Maybe the Middle East.

The Hon. MARK BUTTIGIEG: Or the Middle East.

The CHAIR: I've been a bit obsessed throughout this inquiry about what is critical when it comes to data centres. We hear this line—particularly from the data centre industry—that data centres are critical infrastructure, that we can't do without them, they're just the same as any other essential service. But what's becoming clear is that there are a multitude of different uses and things that are going on in these data centres, everything from storing government data and health data, things that that help the financial sector work et cetera et cetera, all the way down to your cat videos and your Trump as Jesus photos. From your research with communities, do you think that, if we could explain exactly what a data centre was being used for, you would then get better community engagement about is it worth it in our community to have this thing right next to a school? Do you think that there would be greater buy-in if people thought it was genuinely for something critical as opposed to something frivolous?

SANAA SHAH: I think for sure. Something, again, that's happening in community conversations that we're having, especially with young folks, is just frustration that it is the cat videos, it is generative AI. It's a boom that's driving the demand in these really large data centres. I think being able to explain what type of AI or what the purposes are would help. It's something that people do want to be doing research into. Yes, there is just a lot of frustration that our environments are potentially being destroyed just to create those silly cat videos. That's definitely a very common sentiment.

EMMA BACON: Yes. I think there's maybe pressure from the industry to say that this must be done quickly; otherwise there'll be costs. But, in fact, there are costs to doing it quickly. I think that when it comes to both concern for people who interact with AI, who don't necessarily think that not only do they have to deal with AI slop on their phone and that their kids have to interact with, and things like that, but also they have to look at the infrastructure that delivers it. Those things are very much linked for people, I think.

SANAA SHAH: I know the inquiry is about data centres, but you can't separate AI and the generative AI and the harms.

The CHAIR: That's right. There's a difference between data centres for—on the technicalities, Ms Munro knows more than me, but there's the cloud use and then there's the stuff that's more for generative AI and these hyperscale data centres. I think that physical manifestation of a thing that a lot of people have a little bit of antipathy towards, it does—

The Hon. JACQUI MUNRO: The cloud was so ephemeral, but data centres are so physical.

The Hon. MARK BUTTIGIEG: Oh, hang on, it's not in outer space; it's here.

The CHAIR: Yes, that's right. It's here and it's right near my school, and I'm also really annoyed every time I get an AI prompt come up. There does seem to be that connection of people just going, "Why are we doing this for this?"

EMMA BACON: Yes, I think that's something that we've observed. I also say that this is linked to a broader concern by lots of communities in Western Sydney that they are not getting the infrastructure they're asking for. Public transport, parks, trees, swimmable rivers, good schools, libraries, community centres, pools—things that people really want are a low priority, or they perceive them as a low priority. They perceive their community's concerns as being a low priority. When things are rushed through, they say, "We don't know the environmental impact. We don't know the energy impact. We don't know what it's going to do to the air quality next to the school." This broadly undermines trust in government. It's not just about technology and AI; it's actually about how people perceive not just this issue but many others where they might interact with the government and question who is serving who.

The CHAIR: Very well said. I think that even if it didn't have all of the other issues, just having construction on your street can be quite an inconvenience. If you were suffering through a period of inconvenience for something like a community swimming pool or a library or a major bit of transport infrastructure, then you're more likely to be accepting of that than if you haven't had any of those things and now it's a data centre. Thank you so much. This has been very useful. I'm sure there will be supplementary questions, and the Committee secretariat will be in touch about those. That does conclude our time with you, and it concludes our hearing for today.

(The witnesses withdrew.)

The Committee adjourned at 14:50.